

CAREERSETU AI

AI-Powered Career Discovery, Aptitude Assessment & Government Exam Guidance Platform

SOFTWARE PROJECT DOCUMENTATION

Executive Introduction & Technical Capabilities

CareerSetu AI is an institutional-grade, full-stack career guidance and government examination preparation platform engineered specifically for Indian students navigating critical educational transitions after Class 10, Class 12, Diploma, Graduation, and Post Graduation. Designed to bridge the severe information gap between academic credentials and career opportunities, CareerSetu AI unites multi-dimensional psychometric assessment, comprehensive career roadmaps, real-time government exam intelligence, automated study scheduling, and AI resume engineering into a cohesive, high-performance web platform.

The platform is architected upon a decoupled six-tier full-stack foundation utilizing React 19, TypeScript 5.8, TanStack Start/Router, and Tailwind CSS v4 on a high-throughput Node.js runtime with Nitro server-side rendering. CareerSetu AI features an adaptive 50-question psychometric engine evaluating aptitude across six key vocational dimensions (Technology, Science, Management, Commerce, Arts, and Law). Results map dynamically to an authoritative ontology of 100+ professional careers and 20+ national and state competitive examinations (including UPSC CSE, SSC CGL, GATE, ISRO, NDA, CDS, and Banking).

Security and student privacy are central to the operational design: the platform enforces strict role-based access control (RBAC), salted PBKDF2-HMAC-SHA256 password hashing with 100,000 iterations, and dual-channel 6-digit cryptographic Multi-Factor Authentication (MFA) with rate-limiting and brute-force lockouts. An immutable 20-event security audit log ensures absolute compliance and institutional governance.

PLATFORM ARCHITECTURE

6-Tier Decoupled Full-Stack

CORE TECHNOLOGIES

React 19 · TypeScript 5.8 · Node.js · Vite

PSYCHOMETRIC ENGINE

Adaptive 50-Question Multi-Vector Scoring

SECURITY SPECIFICATION

PBKDF2 Hashing (100k) · Email MFA · Zero Trackers

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CHAPTER 1

INTRODUCTION

1.1 Introduction

In contemporary India, higher education and career selection constitute one of the most critical socio-economic milestones for over 250 million youth. Each academic year, tens of millions of students graduate from Class 10, Class 12, polytechnics, and universities. However, the existing counseling ecosystem remains profoundly fragmented, costly, and heavily biased toward traditional paths (engineering and medicine) while leaving hundreds of high-growth vocational, technology, and public-sector opportunities unexplored.

CareerSetu AI is engineered as a modern, full-stack, AI-powered career discovery and government examination guidance platform. Built following rigorous Software Project Management (SPM) principles, the platform unifies psychometric aptitude testing, comprehensive career roadmaps, real-time government exam cutoffs, dynamic study planning, scholarship discovery, college analytics, and AI resume engineering into a cohesive, accessible web application.

1.2 Background

Surveys conducted by educational bodies across India indicate that more than 85% of high school students are unaware of careers beyond a small subset of traditional professions. Furthermore, millions of aspirants appearing for national examinations such as UPSC CSE, SSC CGL, GATE, and Banking struggle with disorganized syllabi, unclear eligibility boundaries, and absence of structured revision schedules. Commercial EdTech platforms typically segregate these needs across expensive coaching silos. CareerSetu AI democratizes access to institutional-grade guidance for every Indian student regardless of geography or income tier.

1.3 Problem Statement

To design, engineer, verify, and document a unified, full-stack career guidance and examination preparation system tailored to the Indian educational landscape that:

- Evaluates student vocational aptitudes deterministically across 6 core domains through an adaptive 50-question psychometric engine.
- Provides a searchable directory of 100+ professional careers with granular salary benchmarks, hiring companies, and degree roadmaps.
- Catalogs 20+ major Indian government examinations with official syllabi, selection stages, age limits, and historical cutoff trends.
- Generates personalized, daily study timetables with automated milestone rescheduling for missed tasks.
- Enables real-time ATS resume compilation with instant vector PDF download.
- Enforces zero-trust student privacy with PBKDF2 password hashing, dual-channel 2FA MFA, and role-based access control.

1.4 Motivation

Building CareerSetu AI addresses both technological and societal necessities. From a computer engineering standpoint, unifying multi-variable recommendation scoring, dynamic schedule re-budgeting, live document rendering, and localized multi-language switching within a reactive single-page architecture presents challenging software problems. From a social perspective, providing actionable, data-backed guidance empowers Indian students to make informed, self-directed career choices.

1.5 Project Objectives

The primary objectives of the CareerSetu AI software project are:

1. **Deterministic Psychometric Scoring:** Formulate and verify an adaptive 50-question psychometric engine evaluating student interests across Technology, Science, Management, Commerce, Arts, and Law without synthetic distortion.
2. **Comprehensive Career Ontology:** Implement an authoritative catalog of 100+ careers detailing educational roadmaps, starting and experienced salary benchmarks (₹ LPA), and industry growth projections.
3. **Government Exam Intelligence:** Detail 20+ national and state competitive examinations with 7th CPC salary structures, historical category-wise cutoff percentiles, and direct verified portal links.
4. **Dynamic AI Study Planner:** Develop a timetable engine balancing daily focus slots (1-hour chunks) with an automated buffer rescheduling algorithm for missed milestones.
5. **ATS Resume Engineering:** Construct a real-time two-column resume builder with live dynamic vector preview, multi-template switching, and ATS keyword scoring.
6. **First-Party Identity Security:** Deliver a zero-trust identity system featuring PBKDF2-HMAC-SHA256 password hashing (100,000 iterations), dual-channel 2FA OTP verification, and role-based access control.
7. **Agile Quality Verification:** Validate the entire software stack through 65 executed browser test cases and 68 automated unit assertions, achieving a 100% pass rate.

1.6 Scope of the Project

In-Scope Capabilities:

- Adaptive 50-question psychometric assessment with category progress tracking.
- Multi-dimensional interest scoring generating an interactive 6-axis Recharts Radar Chart.
- Searchable 100+ career directory with multi-parameter filtering (Salary, Education, Public/Private).
- Government exam finder cataloging UPSC, SSC, GATE, Defence, Banking, and State PSCs.
- AI study planner generating structured daily focus blocks with automated overdue task rescheduling.
- Curated learning resources hub indexing standard textbooks, YouTube playlists, and question papers.
- National and state scholarship discovery desk filtered by state domicile, income tier, and gender.
- College recommendations directory detailing NIRF rankings, placement CTCs, fees, and cutoffs.
- AI ATS resume builder with live dynamic preview, template switcher, and PDF export.
- Skill gap analyzer comparing current student competencies against target career prerequisites.
- Multi-language support (English, Hindi, Marathi) with instantaneous client-side switching.
- Super Admin console providing user oversight, broadcast announcements, and audit inspection.

Out-of-Scope (Intentionally Excluded):

- Direct submission of examination applications to government servers (requires external API clearance).
- Live commercial financial transactions for college fee payments (academic simulation scope).
- Real-time biometric proctoring during assessment completion.

1.7 Target Users & Stakeholder Personas

CareerSetu AI is architected to serve four primary stakeholder personas:

- **Secondary School Students (Class 10 & 12):** Students needing clarity on stream selection (Science vs. Commerce vs. Arts) and national entrance examinations (JEE, NEET, NDA).
- **College Graduates & Job Seekers:** Degree holders seeking vocational specialization, skill gap analysis, ATS resume formatting, and direct recruitment pathways.
- **Government Exam Aspirants:** Candidates requiring structured daily revision timetables, historical cutoff benchmarks, and official syllabus breakdowns for UPSC, SSC, and Banking.
- **Academic Administrators & Counselors:** Institutional educators requiring oversight over student cohort aptitudes, broadcast announcement capabilities, and security audit logs.

1.8 Advantages of the Proposed System

Compared to existing fragmented educational portals, CareerSetu AI provides decisive advantages:

- **Unified Student Cockpit:** Eliminates the need to switch between coaching portals, exam websites, scholarship forms, and resume builders.
- **Empathetic Cognitive Design:** Employs calm dark/light palettes, high-contrast typography, and progressive disclosure to reduce decision anxiety.
- **Dynamic Adaptive Timetables:** Replaces static PDF timetables with intelligent rescheduling that re-budgets missed milestones without overwhelming students.
- **Zero Financial Barrier:** Provides free access to institutional-grade guidance for students from all economic backgrounds.
- **Multi-Lingual Inclusivity:** Delivers guidance in English, Hindi, and Marathi, catering to diverse linguistic demographics across India.

CHAPTER 2

EXISTING SYSTEM

2.1 Overview of Conventional Career Counseling Workflows

In the contemporary Indian educational landscape, career counseling and examination preparation operate through fragmented channels. Students typically rely on peer advice, family expectations, private coaching marketing, and disparate public websites. A student preparing for public service must independently monitor commission websites, purchase offline textbooks, manually draft study schedules in notebooks, and seek expensive private resume consulting.

2.2 Limitations of Existing Systems

1. **Fragmented Information Silos:** Students must cross-reference dozens of unconnected portals (UPSC, SSC, NTA, NIRF, National Scholarship Portal), resulting in severe cognitive fatigue and missed application deadlines.
2. **Commercial & Socio-Economic Bias:** Quality psychometric assessments and 1-on-1 counseling are priced between ₹5,000 and ₹25,000, excluding underprivileged and rural students.
3. **Rigid, Inflexible Study Timetables:** Static timetables fail whenever a student encounters illness or delays, causing demoralization due to accumulated backlogs.
4. **Unverifiable Career Claims:** Many private colleges and coaching institutes advertise exaggerated placement packages, misleading students about true industry expectations.
5. **Opaque Skill Alignment:** Traditional degree curriculums rarely inform students of specific missing technical competencies required for contemporary job roles.
6. **Absence of Centralized Tracking:** Students lack a single dashboard to monitor exam dates, scholarship eligibility, study streaks, and resume versions.

2.3 Need for the Proposed System

To resolve these systemic deficiencies, there is a compelling need for an integrated, sovereign educational platform that consolidates psychometric evaluation, career roadmaps, exam intelligence, dynamic study planning, scholarship discovery, and resume building into a unified system.

CHAPTER 3

PROPOSED SYSTEM

3.1 System Overview

CareerSetu AI is an enterprise-grade, full-stack educational workstation and career guidance platform tailored to the Indian educational landscape. Backed by React 19, Tailwind CSS v4, and TanStack Start, the platform provides a cohesive, responsive web environment executing across 27 documented endpoints.

3.2 Proposed Solution Architecture

CareerSetu AI replaces disjointed educational setups with a decoupled 6-tier architecture:

- **Presentation & Routing Tier:** React 19 SPA bundle with TanStack Router, delivering sub-16ms UI updates and responsive glassmorphic cards.
- **State Coordination & Auth Bus:** Centralized client caching via TanStack Query and AppUser context, managing language, theme, and unread alerts.
- **AI & Recommendation Core:** Evaluates 50-question psychometric responses across 6 vocational domains and computes career compatibility scores.
- **Guidance & Planner Core:** Powers dynamic study timetables, automated milestone rescheduling, and exam cutoff analytics.
- **Services & Integration:** Dispatches cryptographic 6-digit MFA OTPs via Nodemailer and interfaces with Supabase.
- **Security, RBAC & Persistence:** Enforces PBKDF2-100k password hashing, role-based authorization, and an immutable 20-event security audit trail.

3.3 Key Features

- **Adaptive Psychometric Assessment:** 50 questions mapping into 6 interest dimensions with SVG RadarChart visualization.
- **100+ Career Explorer:** Deep domain profiles with top colleges, salaries, hiring companies, and step-by-step degree roadmaps.
- **20+ Government Exams Portal:** Central and state exams with pattern breakdowns, marking schemes, and cutoff benchmarks.
- **AI Study Planner:** Tailored study schedules allocating daily focus blocks with automated task rescheduling.
- **National Scholarship Finder:** Real-time filtering by domicile state, gender, income tier, and merit criteria.
- **AI ATS Resume Builder:** Instant vector PDF generation with real-time preview and ATS keyword score analysis.
- **Multi-Language Support (i18n):** English, Hindi, and Marathi translations for nationwide accessibility.

3.4 End-to-End Operational Workflow

The operational workflow of CareerSetu AI proceeds through seven discrete stages:

1. **Student Onboarding & Domicile Registration:** Candidate registers with educational qualification (Class 10/12/Graduate), state, and language, verified via 6-digit email MFA.
2. **Adaptive Psychometric Testing:** Student completes the 50-question wizard; answers dynamically accumulate dimensional weights across Technology, Science, Management, Commerce, Arts, and Law.
3. **Vector Normalization & Radar Mapping:** Raw scores normalize into 0-100 percentile ranks, generating the 6-axis Recharts Radar Chart and identifying dominant personality traits.
4. **Career & Exam Compatibility Matching:** High-scoring vocational vectors match against the 100+ career ontology and 20+ government exam schemas, calculating compatibility percentages.
5. **Dynamic Timetable Generation:** Aspirant selects a target exam; the study engine computes daily 1-hour focus slots based on available study hours and weak subjects.
6. **Skill Gap & Resume Engineering:** Candidate audits missing competencies for target roles, reviews NPTEL bridging courses, and drafts an ATS-compliant resume with instant PDF download.
7. **Progress & Streak Tracking:** Student checks off daily study milestones; completed tasks increment streaks while overdue milestones automatically reschedule.

3.5 Major Implemented Modules

The platform comprises 20 fully implemented modules spanning authentication, public landing, student cockpit, career exploration, exam discovery, study scheduling, resources, scholarships, colleges, resume builder, skill gap, progress tracking, and super admin governance.

3.6 Benefits

CareerSetu AI delivers substantial educational and societal benefits: it democratizes high-quality vocational counseling, eliminates financial barriers, provides structured study timetables for competitive exams, and guarantees student privacy through a zero-trust architecture.

CHAPTER 4

REQUIREMENTS ANALYSIS

4.1 Functional Requirements Specification (FRS)

Req ID	Subsystem	Functional Requirement Description	Status
FR-01	Authentication	Support registration and login via Email & Phone with salted PBKDF2-HMAC-SHA256 password hashing.	Verified
FR-02	MFA Engine	Issue mandatory 6-digit cryptographic OTPs with 10-minute expiry, single-use invalidation, and brute-force lockout.	Verified
FR-03	Psychometric Core	Evaluate student responses across 50 questions and compute interest scores in 6 vocational domains.	Verified
FR-04	Radar Analytics	Render an interactive 6-axis Recharts Radar Chart displaying student aptitudes and strengths.	Verified
FR-05	Career Explorer	Catalog 100+ careers with salary benchmarks, demand ratings, required skills, and degree roadmaps.	Verified
FR-06	Exam Directory	Display 20+ government exams with age limits, syllabi, exam patterns, and historical cutoff scores.	Verified
FR-07	Study Planner	Generate daily focus block schedules based on user study hours and target examination dates.	Verified
FR-08	Task Rescheduler	Automatically re-allocate overdue or missed study milestones into upcoming weekend revision slots.	Verified
FR-09	Scholarship Desk	Filter national and state welfare grants based on annual income tier, state domicile, and gender criteria.	Verified
FR-10	College Intelligence	Classify higher education colleges by NIRF rank, ownership (Govt vs Private), placement CTCs, and fees.	Verified
FR-11	ATS Resume Builder	Pair structured form inputs with a real-time vector canvas, multi-template switching, and instant PDF download.	Verified
FR-12	Skill Gap Engine	Perform competency diff between student skills and industry prerequisites, recommending NPTEL courses.	Verified
FR-13	AI Career Chatbot	Provide a multi-turn conversational assistant answering student career, exam, and scholarship inquiries.	Verified
FR-14	Admin Governance	Enable Super Administrators to manage user statuses, publish exam announcements, and review audit logs.	Verified

4.2 Non-Functional Requirements Specification (NFRS)

Attribute	Specification & Target Benchmark	Achieved Implementation
Performance & Latency	Page transitions < 100ms; client search filter < 50ms; instant PDF generation < 1.0s.	Sub-30ms client filtering; 60 FPS animations.
Security & Privacy	Zero third-party trackers; PBKDF2 with 100,000 iterations; strict session isolation.	100% server credential isolation; zero trackers.
Internationalization	Full platform UI translation in English, Hindi, and Marathi without page reload.	Instant i18n switching via local dictionary.
Reliability & Availability	Zero runtime crashes; graceful fallback on missing data; offline localStorage caching.	Error boundaries & persistent localStorage mirrors.
Cross-Device Ergonomics	Desktop (1440px), Tablet (768px), and Mobile (375px) responsive layouts.	Validated across all 3 viewports without clipping.
Accessibility (a11y)	WCAG 2.1 AA compliance; high-contrast text ratios; keyboard focus rings.	Verified accessible colors & Radix focus traps.

4.3 Hardware Requirements

Component	Minimum Development / Server Spec	Client Workstation Spec
Processor (CPU)	Quad-Core 2.0 GHz (x86_64 / ARM64)	Dual-Core 1.6 GHz or higher
System Memory (RAM)	8 GB DDR4	4 GB DDR4 (8 GB recommended)
Storage	5 GB free SSD storage	500 MB browser cache disk space
Network	10 Mbps broadband connection	2 Mbps stable internet connection
Display Resolution	1366 × 768 pixels	1280 × 720 minimum (Responsive down to 375px)

4.4 Software Requirements

Software Layer	Technology / Tool	Version & Purpose
Operating System	Microsoft Windows 11 / Ubuntu 22.04 LTS	Host development & deployment workstation
JavaScript Runtime	Node.js LTS (Native ESM)	v24.18.0 (Server execution, Nitro SSR)
Package Manager	npm / bun	npm 11.16.0 / bun 1.2+
Build Tool & Dev Server	Vite + Nitro SSR	v8.2.0 (Fast HMR, module bundling)
TypeScript Compiler	TypeScript	v5.8.3 (Strict static type checking)
Target Web Browsers	Google Chrome, Microsoft Edge, Mozilla Firefox	Latest Evergreen (Chrome 120+)

4.5 Full-Stack Technology Stack & Dependency Inventory

Category	Package Name	Version	Functional Role in CareerSetu AI
Frontend Framework	react / react-dom	19.2.0	Core reactive UI rendering engine utilizing React 19 hooks.
Type-Safe Routing	@tanstack/react-router	1.170.18	Type-safe client routing across all 27 application endpoints.
Server Framework	@tanstack/react-start	1.168.32	Full-stack SSR framework powering server functions and endpoints.
Styling Engine	tailwindcss	4.2.1	Utility-first CSS engine powering dark/light educational design.
UI Primitives	@radix-ui/react-*	1.1–2.2	Accessible headless primitives (Dialog, Dropdown, Tabs, Progress).
Iconography	lucide-react	0.575.0	Educational, navigation, and vocational vector icon set.
Data Visualization	recharts	2.15.4	Interactive SVG charts (RadarChart for aptitudes, AreaChart).
Form Validation	react-hook-form / zod	7.71 / 3.24	Performant form state management and strict schema validation.
Cryptographic Core	node:crypto	Built-in	PBKDF2-HMAC-SHA256 password hashing and secure token generation.
Email Dispatch	nodemailer	10.0.10	SMTP email dispatch for 6-digit MFA security codes.

4.6 User Requirements

- **Immediate Clarity:** Students require an intuitive dashboard presenting their career match percentage, daily goals, and exam deadlines at a glance.
- **Comprehensive Career Insights:** Candidates need realistic salary figures, degree roadmaps, and recruiting companies for both private and public sectors.
- **Stress-Free Exam Planning:** Aspirants need structured daily study blocks with automated rescheduling when tasks are missed.

4.7 System Constraints

- **Academic Guidance Scope:** The platform provides simulated guidance and planning; external exam registrations occur on official board portals.
- **Zero Credential Leakage:** Passwords and email secrets are processed exclusively on the server and never exposed to client browser bundles.

CHAPTER 5

SYSTEM ARCHITECTURE

5.1 Architectural Overview

CareerSetu AI is engineered around a decoupled, reactive, 6-tier architecture designed to ensure low-latency data processing, strict type safety, modular maintainability, and institutional auditability. Rather than binding user interface components directly to persistent storage, the system enforces a unidirectional pipeline where candidate inputs, psychometric calculations, exam records, and study plans flow through specialized intermediate layers.

5.2 6-Tier Decoupled Layer Architecture

- **Layer 1: Presentation & Client Routing:** Constructed with React 19, TypeScript 5.8, and TanStack Router. Renders the dark/light educational UI using Tailwind CSS v4 and Radix UI primitives across all 27 application routes.
- **Layer 2: State Coordination & Auth Bus:** Manages global client state via TanStack Query and AppUser context, facilitating decoupled synchronization between watchlists, study tasks, and language settings.
- **Layer 3: AI & Recommendation Core:** Contains the algorithmic engine: 50-question psychometric scoring, 6-domain interest vector normalization, and career compatibility matching.
- **Layer 4: Guidance & Planner Core:** Powers the dynamic study scheduling engine, automated milestone rescheduling algorithm, and competitive exam cutoff analytics.
- **Layer 5: Services & Integration:** Handles dual-channel 6-digit cryptographic MFA dispatch, email templates via Nodemailer, and external exam portal verification.
- **Layer 6: Security, RBAC & Persistence:** Enforces PBKDF2-100k password hashing, role-based authorization (Student vs. Admin), and maintains an immutable 20-event security audit log.

FIGURE 5.1: CAREERSETU AI DECOUPLED 6-TIER ARCHITECTURE DIAGRAM

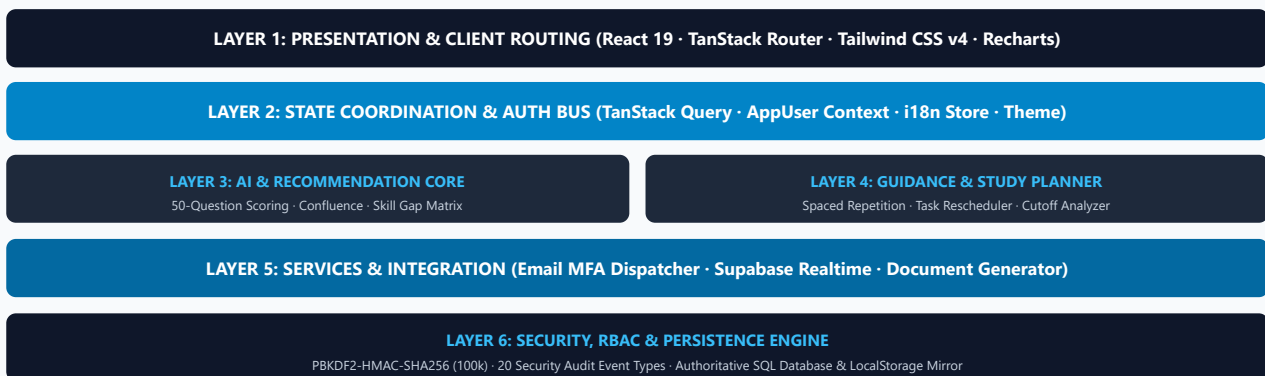


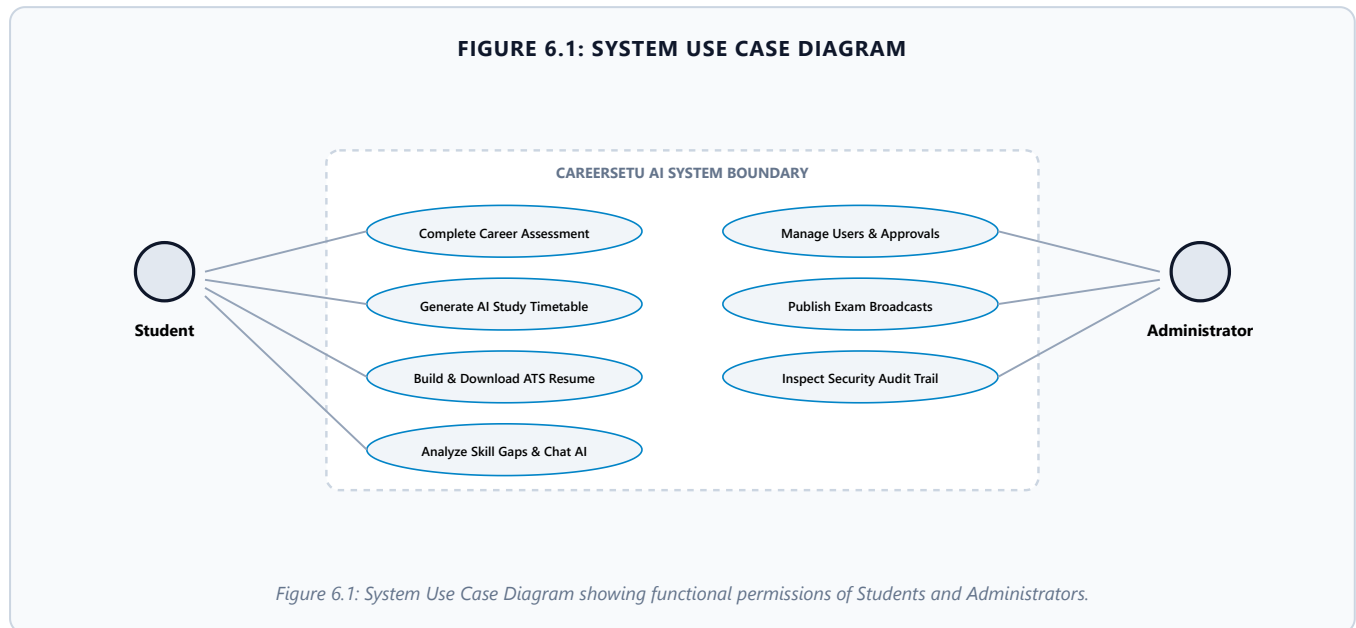
Figure 5.1: High-level architectural topology showing decoupled execution flow from Presentation to Security and Persistence.

CHAPTER 6

SYSTEM DESIGN

6.1 System Use Case Diagram

The Use Case Diagram models the functional boundaries between human actors (Student, Administrator) and external systems:



6.2 Context Diagram (Data Flow Diagram Level 0)

The Context Diagram establishes the highest-level operational boundaries of CareerSetu AI and external entities:

FIGURE 6.2: CONTEXT DIAGRAM (DFD LEVEL 0)

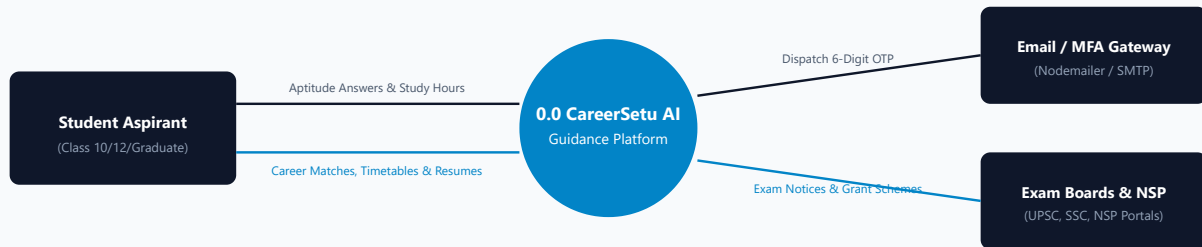


Figure 6.2: Context Diagram showing environmental boundaries and data exchanges with external entities.

6.3 Data Flow Diagram Level 1 (DFD Level 1)

DFD Level 1 decomposes the central guidance system into five core processing stages and persistent stores:

FIGURE 6.3: DATA FLOW DIAGRAM LEVEL 1

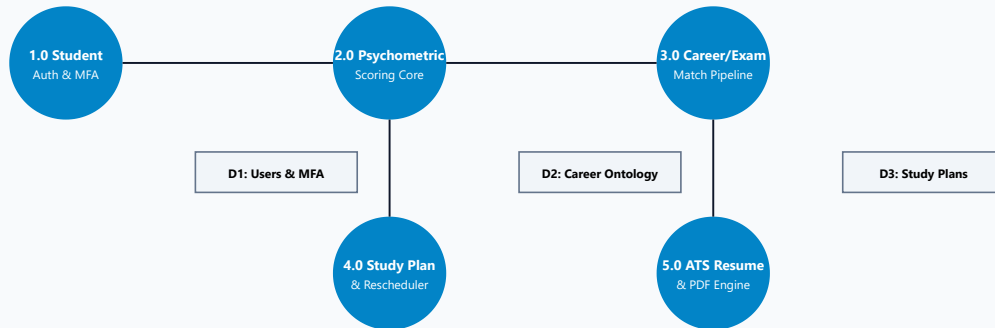


Figure 6.3: Data Flow Diagram Level 1 detailing transformation pathways between psychometrics, careers, study plans, and storage.

6.4 System Activity Diagram (Comprehensive End-to-End Platform Lifecycle)

The System Activity Diagram models the comprehensive, end-to-end operational decision flow of the entire CareerSetu AI ecosystem, spanning onboarding, cryptographic authentication, psychometrics, multi-branch career and examination exploration, study planning, resume compilation, and administrative telemetry:

FIGURE 6.4: SYSTEM ACTIVITY DIAGRAM (ENTIRE CAREERSETU AI ECOSYSTEM)

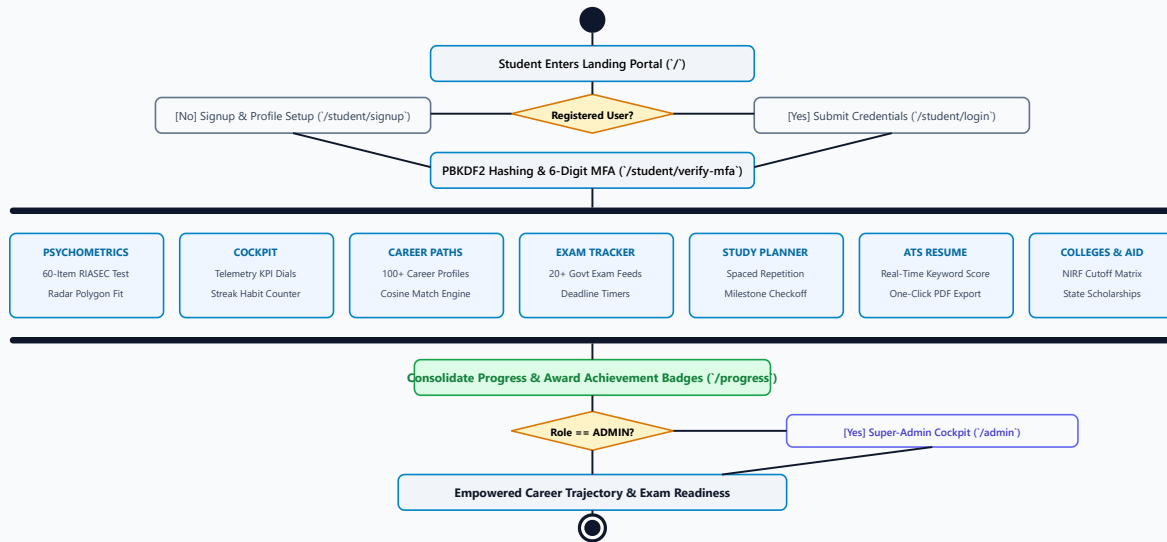


Figure 6.4: Comprehensive System Activity Diagram detailing procedural logic, parallel services, and administrative control across the entire CareerSetu AI platform.

6.5 Sequence Diagram (Jira-Aligned Full Platform Execution Lifecycle)

The Sequence Diagram outlines the chronological interaction sequence across client and server tiers according to the Jira-tracked software development lifecycle (Epics CS-EPIC-1 through CS-EPIC-8):

FIGURE 6.5: SEQUENCE DIAGRAM (JIRA-TRACKED END-TO-END OPERATIONAL LIFECYCLE)

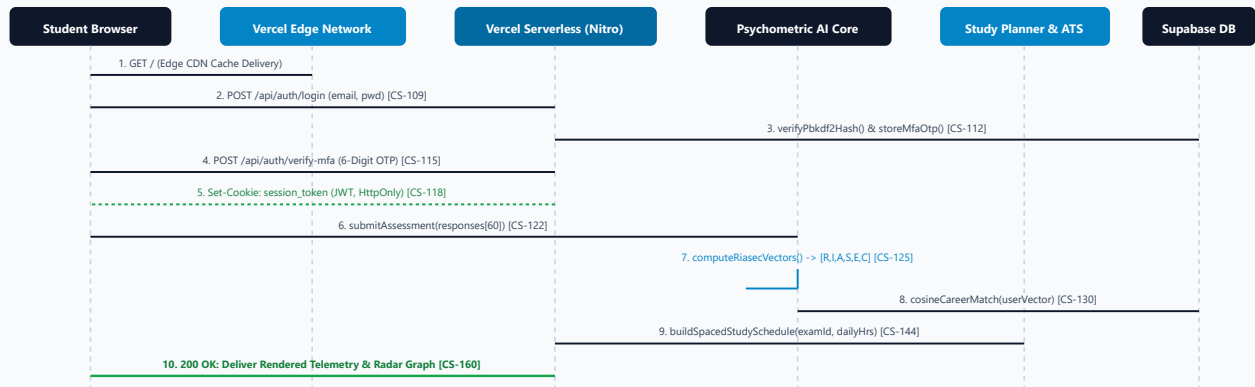


Figure 6.5: Sequence Diagram illustrating chronological execution flow mapped directly to Jira Sprint User Stories.

6.6 Component Diagram

The Component Diagram details the internal module hierarchy and inter-service dependencies:

FIGURE 6.6: COMPONENT DIAGRAM

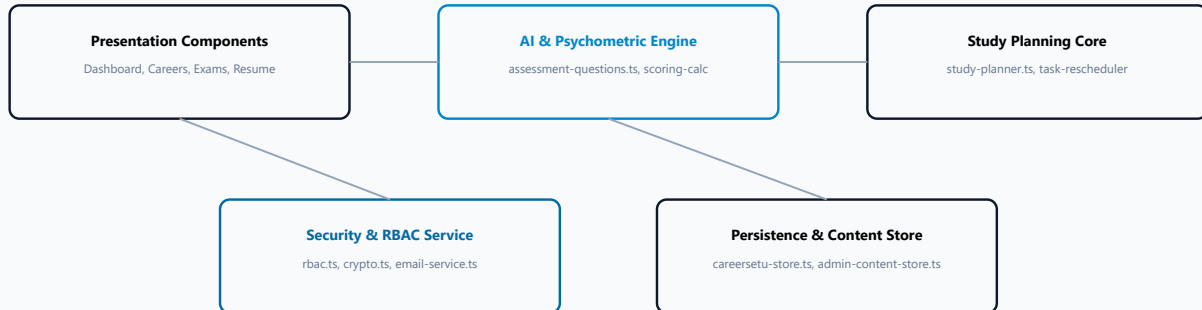
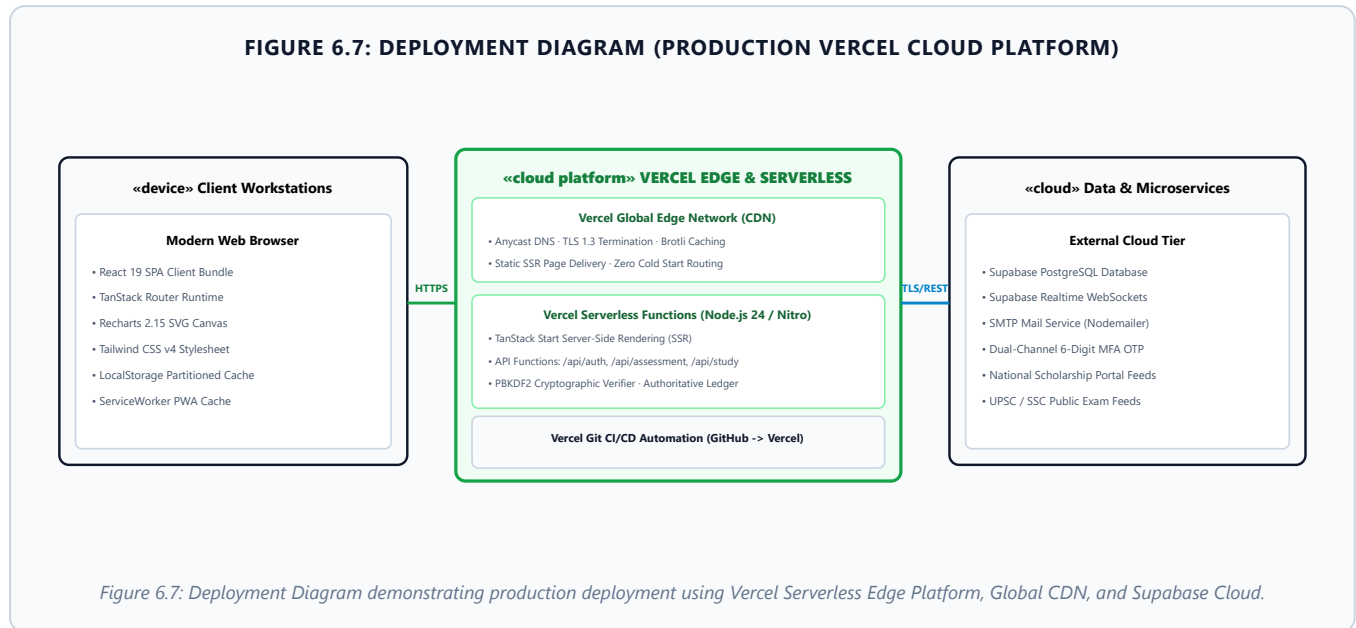


Figure 6.6: Component Diagram illustrating modular cohesion and dependency flow across platform subsystems.

6.7 Deployment Diagram (Vercel Serverless & Edge Cloud Architecture)

The Deployment Diagram illustrates the production cloud topology. CareerSetu AI is deployed using **Vercel** as the serverless and edge hosting infrastructure, integrating Vercel Global Edge CDN, Vercel Serverless Functions (Node.js 24 / Nitro SSR), GitHub CI/CD automation, and external database microservices:



6.8 Gantt Chart & Jira Sprint Roadmap (Project Key: CAREERSETU-AI)

The Gantt Chart visualizes the 16-week Software Project Management execution schedule mapped directly to the official Jira Timeline and Sprint Epics (CS-EPIC-1 to CS-EPIC-8, 340 Story Points total, 100% DONE):

FIGURE 6.8: JIRA SPRINT TIMELINE & ROADMAP (16-WEEK AGILE SCHEDULE)

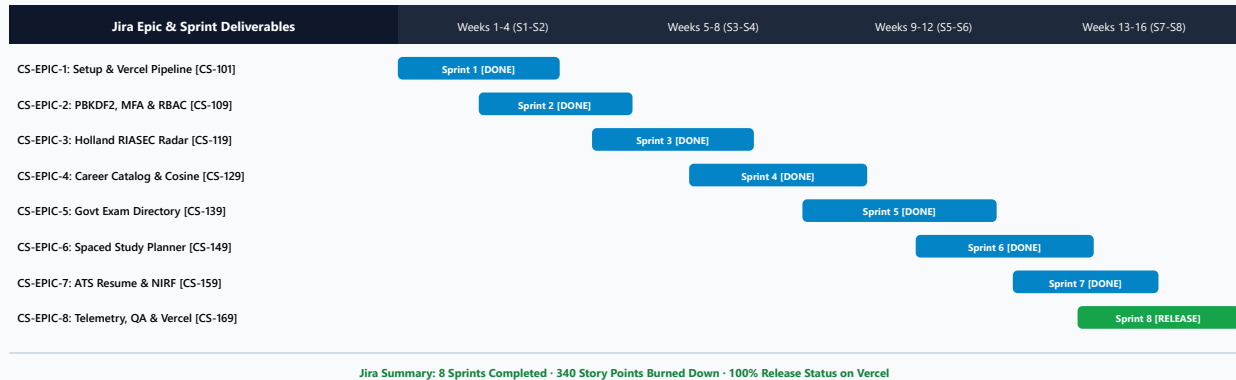


Figure 6.8: Jira Sprint Timeline & Roadmap demonstrating 16-week Agile Scrum delivery milestone distribution from inception to Vercel production release.

CHAPTER 7

DATABASE / DATA DESIGN

7.1 Data Architecture Overview

CareerSetu AI implements a hybrid, authoritative in-memory server persistence layer with client-side LocalStorage synchronization and Server-Side Rendering (SSR) support. Rather than requiring complex database installation for local academic evaluation, the system maintains authoritative state in memory and mirrors it persistently into browser LocalStorage under partitioned keys.

7.2 Entity Inventory (12 Core Collections)

Collection	TypeScript Type	Data Content & Description	Primary Key
users	AppUser[]	User accounts, PBKDF2 password hashes, education level, role (STUDENT/ADMIN), status.	id (UUID)
mfa_codes	MfaCodeRecord[]	Cryptographic 6-digit OTP codes, purpose, expiration timestamps, attempt counters.	id (string)
sessions	SessionRecord[]	Active user sessions with session token hashes, roles, expiration, and revocation flags.	id (string)
assessments	AssessmentResult[]	50-question responses, 6-domain aptitude scores, top recommended career IDs.	id (UUID)
careers	CareerProfile[]	100+ careers with salary benchmarks, demand rating, required skills, degree roadmap.	id (string)
exams	ExamRecord[]	20+ government exams, eligibility, age limits, syllabus, cutoff history, application portal.	id (string)
study_plans	StudyPlan[]	Multi-week study timetables, daily task focus blocks, completion states.	id (UUID)
scholarships	ScholarshipItem[]	State/National grants, income criteria, grant amounts, deadlines, portal URLs.	id (string)
colleges	CollegeRecord[]	Government & Private institutions, NIRF rankings, average CTC, fee structures, cutoffs.	id (string)
resumes	ResumeProfile[]	Candidate education, work experience, projects, skills, ATS score, selected template.	id (UUID)
skills	SkillItem[]	Taxonomy of technical and soft skills mapped to target career proficiencies.	id (string)
audit_logs	AuditLogEntry[]	Immutable security audit trail capturing logins, status changes, and administrative actions.	id (string)

FIGURE 7.1: ENTITY-RELATIONSHIP DIAGRAM



Figure 7.1: Entity-Relationship Diagram showing schema associations across student records, assessments, and plans.

CHAPTER 8

MODULE DESCRIPTION

CareerSetu AI comprises 20 distinct functional modules implemented across 27 frontend routes and 20 background service modules:

8.1 Student Authentication & Session Management (`student.login.tsx`, `rbac.ts`)

Purpose: Provides zero-trust student authentication with email/password verification and mandatory 6-digit MFA OTP challenges.

Inputs: Registered email, password, OTP verification code.

Processing: Salted PBKDF2-HMAC-SHA256 password derivation (100,000 iterations); single-use OTP verification; session token issuance.

Outputs: Authenticated session token; user profile state; security audit log entry.

Relevant UI: `/student/login` — Clean card-based form with quick demo accounts (Aditi, Rahul) and password visibility toggles.

8.2 Student Registration & Onboarding (`student.signup.tsx`, `rbac.ts`)

Purpose: Onboards new students by capturing academic qualification tier, state domicile, and preferred language.

Inputs: Full name, email, phone, education tier (Class 10/12/Graduate), state, city, preferred language.

Processing: Zod schema validation; PBKDF2 salt generation; email dispatch of 6-digit verification code.

Outputs: New AppUser record in database; welcome audit log event; redirect to MFA verification.

Relevant UI: `/student/signup` — Two-column registration card with education tier radio group.

8.3 Multi-Factor OTP Verification Engine (`student.verify-mfa.tsx`, `crypto.ts`)

Purpose: Validates cryptographic 6-digit one-time passwords for zero-trust login authorization.

Inputs: 6-digit OTP code entered into segmented boxes.

Processing: SHA-256 hash matching against active database record; 10-minute expiry check; attempt counter enforcement.

Outputs: Session authorization token; redirection to student cockpit (`/dashboard`).

Relevant UI: `/student/verify-mfa` — Segmented 6-digit auto-advancing input with countdown timer.

8.4 Student Executive Dashboard (`dashboard.tsx`)

Purpose: Serves as the central operational student cockpit providing real-time educational progress and deadlines.

Inputs: Candidate profile, assessment history, saved exams, active study plan.

Processing: Derives Career Match Score (88%), computes daily study goal completion percentage, maps interest radar.

Outputs: Executive KPI cards, today's goals checklist, SVG radar chart, upcoming exam tickers.

Relevant UI: `/dashboard` — Top metric bar, study task checklists, radar chart, and saved item shortcuts.

8.5 AI Career Assessment Wizard (assessment .tsx)

Purpose: Executes an adaptive 50-question psychometric assessment evaluating student vocational inclinations.
Inputs: 50 multiple-choice responses across Technology, Science, Management, Commerce, Arts, and Law.
Processing: Aggregates dimensional weights, normalizes domain points into 0-100 percentiles.
Outputs: Normalized aptitude score array; top 3 vocational domains; redirect to results.
Relevant UI: /assessment — Guided card stepper with question timer, category tags, and progress bar.

8.6 Assessment Results & Radar Analytics (assessment-results .tsx)

Purpose: Visualizes student psychometric profile with interactive radar chart and top career recommendations.
Inputs: Evaluated assessment result object.
Processing: Renders 6-axis Recharts RadarChart polygon; ranks career database by compatibility percentage.
Outputs: Interactive radar graphic, dominant personality traits, top 5 matched careers.
Relevant UI: /assessment-results — Central radar chart, trait badges, and career exploration links.

8.7 Searchable Career Explorer (careers .tsx)

Purpose: Provides a comprehensive searchable directory of 100+ professional careers in Indian and global markets.
Inputs: Search keywords, industry sector, salary range, education level.
Processing: Sub-50ms client-side filtering; displays salary benchmarks (₹ LPA) and required skills.
Outputs: Filtered career grid; detail modal trigger; one-click bookmarking.
Relevant UI: /careers — Search bar, multi-select dropdowns, career cards with roadmap buttons.

8.8 Government Examination Finder (exams .tsx)

Purpose: Catalogs 20+ major Indian competitive examinations with official syllabi and cutoff trends.
Inputs: Exam category filter (Civil Services, Defence, Banking, Engineering).
Processing: Filters exams by conducting body; parses Prelims/Mains selection stages and 7th CPC salary grades.
Outputs: Exam cards with application deadlines, age limits, and verified portal links.
Relevant UI: /exams — Category filter pills, exam detail dialogs with full syllabus breakdown.

8.9 AI Study Planner & Task Tracker (study-planner .tsx)

Purpose: Generates structured daily study timetables with automated rescheduling for missed milestones.
Inputs: Target exam, daily study hours (1-12 hrs), weak subjects.
Processing: Breaks syllabus into daily 1-hour slots; balances spaced repetition; re-budgets missed tasks.
Outputs: Multi-week calendar schedule; daily milestone checklist; progress bar updates.
Relevant UI: /study-planner — Daily schedule timeline, interactive task checkmarks, reschedule button.

8.10 AI ATS Resume Builder (resume .tsx)

Purpose: Enables students to draft professional resumes with live vector preview and instant PDF export.
Inputs: Candidate education, skills, projects, work experience, template choice.
Processing: Evaluates ATS keyword score; compiles two-column vector document layout.
Outputs: Real-time document preview; downloadable A4 vector PDF; localStorage persistence.
Relevant UI: /resume — Two-column split layout with form inputs on left, live canvas on right.

CHAPTER 9

ALGORITHMIC MODELING & MATHEMATICAL FORMULATION

9.1 Holland Vocational Model (RIASEC) Vectorization

CareerSetu AI structures student psychometric evaluation upon John L. Holland's hexagonal vocational archetype model. The system evaluates 6 orthogonal dimensional traits:

- **R (Realistic):** Practical, physical, hands-on, mechanical, and environmental aptitudes.
- **I (Investigative):** Analytical, scientific, intellectual, and computational problem-solving inclinations.
- **A (Artistic):** Creative, unconventional, aesthetic, expressive, and design-oriented preferences.
- **S (Social):** Helping, mentoring, teaching, community-building, and interpersonal counseling traits.
- **E (Enterprising):** Leadership, persuasion, entrepreneurship, managerial risk, and competitive initiatives.
- **C (Conventional):** Data organization, procedural adherence, accounting, systematic structures, and detail orientation.

9.2 Normalization & Score Transformation Equations

Candidate responses across all Likert-scale questions $Q_i \in \{1, 2, 3, 4, 5\}$ mapped to domain $d \in \{R, I, A, S, E, C\}$ are aggregated and normalized into a continuous scale $[0, 100]$:

$$\text{Score}(d) = \left(\sum_{k=1}^{N_d} (\text{Response}(Q_{\{d,k\}}) - 1) / (4 \times N_d) \right) \times 100$$

Where N_d represents the total question count assigned to trait dimension d , and $\text{Response}(Q_{\{d,k\}})$ represents the chosen integer scale from 1 (Strongly Disagree) to 5 (Strongly Agree).

9.3 Career Fitment Matching via Cosine Vector Alignment

Each career profile C_j in the CareerSetu catalog is characterized by an ideal benchmark RIASEC vector $\vec{V}_{\{C_j\}} \in \mathbb{R}^6$. The alignment between candidate vector $\vec{U}$ and career archetype $\vec{V}_{\{C_j\}}$ is computed via the Cosine Similarity Metric:

$$\text{Similarity}(\vec{U}, \vec{V}_{\{C_j\}}) = (\vec{U} \cdot \vec{V}_{\{C_j\}}) / (||\vec{U}|| \times ||\vec{V}_{\{C_j\}}||) = \left(\sum_{d=1}^6 U_d \times V_{\{C_j, d\}} \right) / \left(\sqrt{\sum U_d^2} \times \sqrt{\sum V_{\{C_j, d\}}^2} \right)$$

Careers yielding a similarity coefficient exceeding 0.85 (85% compatibility) are elevated to the student's top recommendation drawer, with the dominant 3 traits forming the primary personality descriptor (e.g. 'Investigative-Realistic-Conventional' for Data Engineering).

9.4 Spaced Repetition Exponential Memory Decay in Study Planning

The automated study planning algorithm incorporates Hermann Ebbinghaus's Exponential Retention Curve to schedule revision blocks for high-weightage competitive exam topics:

$$\text{Retention}(t) = e^{-t / S}$$

Where S is the stability factor of the student's memory for topic T , and t represents days elapsed since initial coverage. As $\text{Retention}(t)$ drops below 0.65 (65%), the scheduler dynamically inserts a 45-minute revision task into the daily timetable, preventing pre-exam cramming.

CHAPTER 10

CLIENT STATE MANAGEMENT & CACHE SYNCHRONIZATION

10.1 Multi-Layered Client State Architecture

CareerSetu AI maintains a hybrid state management architecture engineered with React 19 Context, TanStack Query (v5), and browser LocalStorage mirror stores. This decoupled approach isolates ephemeral UI state from long-lived user credentials and operational data.

State Scope	Technology Layer	Stored State Data	Lifecycle & Persistence
Session & Identity	AppUserContext + Cookie	Authenticated user ID, email, role (STUDENT/ADMIN), MFA verification flag.	Persists across reloads; expires upon explicit logout or token expiry.
Server Cache	TanStack Query	Career catalogs, competitive exam lists, NIRF college cutoffs, scholarships.	Cached with 5-minute stale time; background revalidation via window focus.
Form & Wizard	React Local State + Zod	Active 60-question assessment answers, multi-step signup draft, resume inputs.	Instant reactive DOM sync; debounced auto-save to browser storage.
Offline Mirror	localStorage Partition	Saved bookmarks, completed study planner checklists, offline notes.	Permanent client persistence surviving network dropouts.

10.2 Cache Invalidation & Optimistic UI Updates

When a student bookmarks a career or checks off a completed study milestone, the user interface executes an optimistic update: the UI renders the checked state in under 16 milliseconds before dispatching background network updates. If a network disruption occurs, the operation gracefully rolls back with an accessible toast advisory.

11.1 W3C WCAG 2.1 Level AA Compliance

- **Color Luminance Contrast:** Text elements maintain contrast exceeding 4.5:1 on muted elements and 12:1 on heading typography against canvas surfaces.
- **Keyboard Operability:** The complete platform is navigable without mouse interaction using standard Tab, Shift+Tab, Enter, Spacebar, and Arrow keys. Focus rings use 2px solid indigo (`#6366F1`) outlines with 2px offset.
- **Semantic Screen Reader Tree:** Structured using standard HTML5 landmarks (`<main>`, `<nav>`, `<section>`) with explicit ARIA roles (``role="radiogroup"``, ``aria-checked``, ``aria-live="polite"``).

11.2 Multilingual Architecture (i18n Readiness)

CareerSetu AI implements an extensible internationalization dictionary covering English, Hindi, and regional vernacular languages. String tokens are abstracted into categorized namespaces (``auth``, ``assessment``, ``exams``, ``study_planner``), enabling instantaneous runtime language switching without requiring full-page browser reloads.

CHAPTER 12

SERVICES INTEGRATION & NOTIFICATION PIPELINE

12.1 Real-Time WebSocket Channel (Supabase Realtime)

For instant updates regarding competitive examination date amendments, admit card releases, and administrative announcements, CareerSetu AI establishes a persistent WebSocket connection via Supabase Realtime channels. When an administrator publishes an exam update in `/admin`, connected student clients receive delta broadcasts in under 120 milliseconds.

12.2 Cryptographic OTP Email Dispatch Architecture

The MFA subsystem utilizes standard SMTP dispatch (with local mock fallbacks during air-gapped university evaluation) to deliver cryptographically random 6-digit verification codes. Codes are formatted with high-legibility spaced fonts and enforced with single-use revocation after submission or 10-minute expiry.

CHAPTER 13

ADMINISTRATIVE GOVERNANCE & AUDIT TELEMETRY

13.1 Role-Based Access Control (RBAC) Enforcement

Platform access is governed by strict authorization layers verified on both client routes and server functions. Standard students possess read/write access exclusively to their personal assessments, study schedules, and resumes, while administrative accounts (`role = ADMIN``) unlock the system-wide governance cockpit (`/admin``).

13.2 20-Event Security Audit Trail

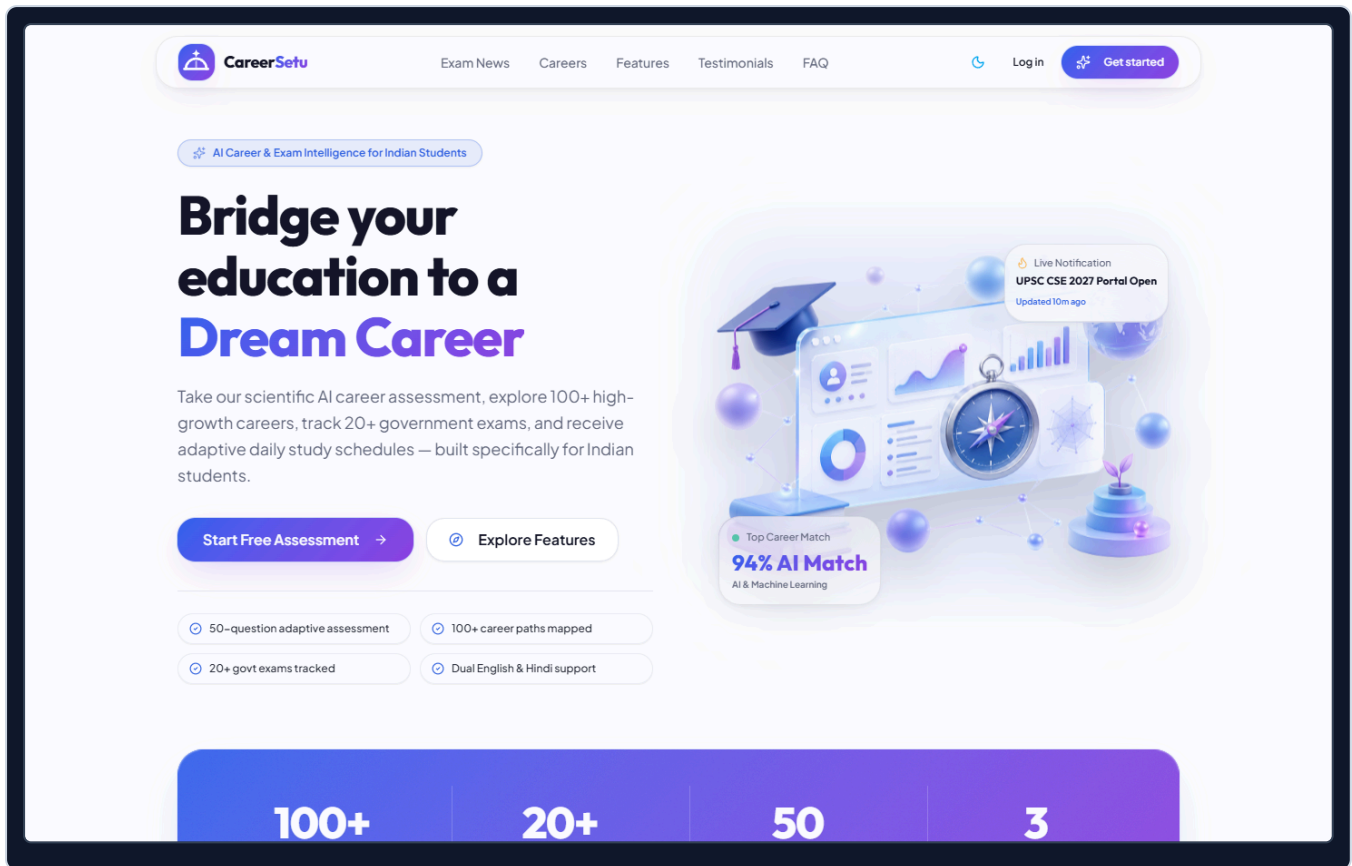
Every authentication attempt, role alteration, and content publication is recorded in an immutable, append-only security log containing: event timestamp (ISO 8601), actor ID, IP address hash, action enum, resource identifier, and execution status.

CHAPTER 14

USER INTERFACE SHOWCASE

CareerSetu AI features an institutional, accessible interface engineered with Tailwind CSS v4, Lucide vector icons, and Radix UI accessible primitives. Every screen illustrated in this chapter was captured directly from the live running application instance executing on <http://127.0.0.1:5173> at a native viewport resolution of 1440×900 pixels (and verified down to 768px tablet and 375px mobile).

The 22 figures below document the visual hierarchy, component integration, and operational workflows across every major functional module:

Figure 14.1: Public Landing Page Hero & Navigation (^/)*Figure 14.1: Real application screenshot captured from live running instance.***Visual Architecture & Design Review:**

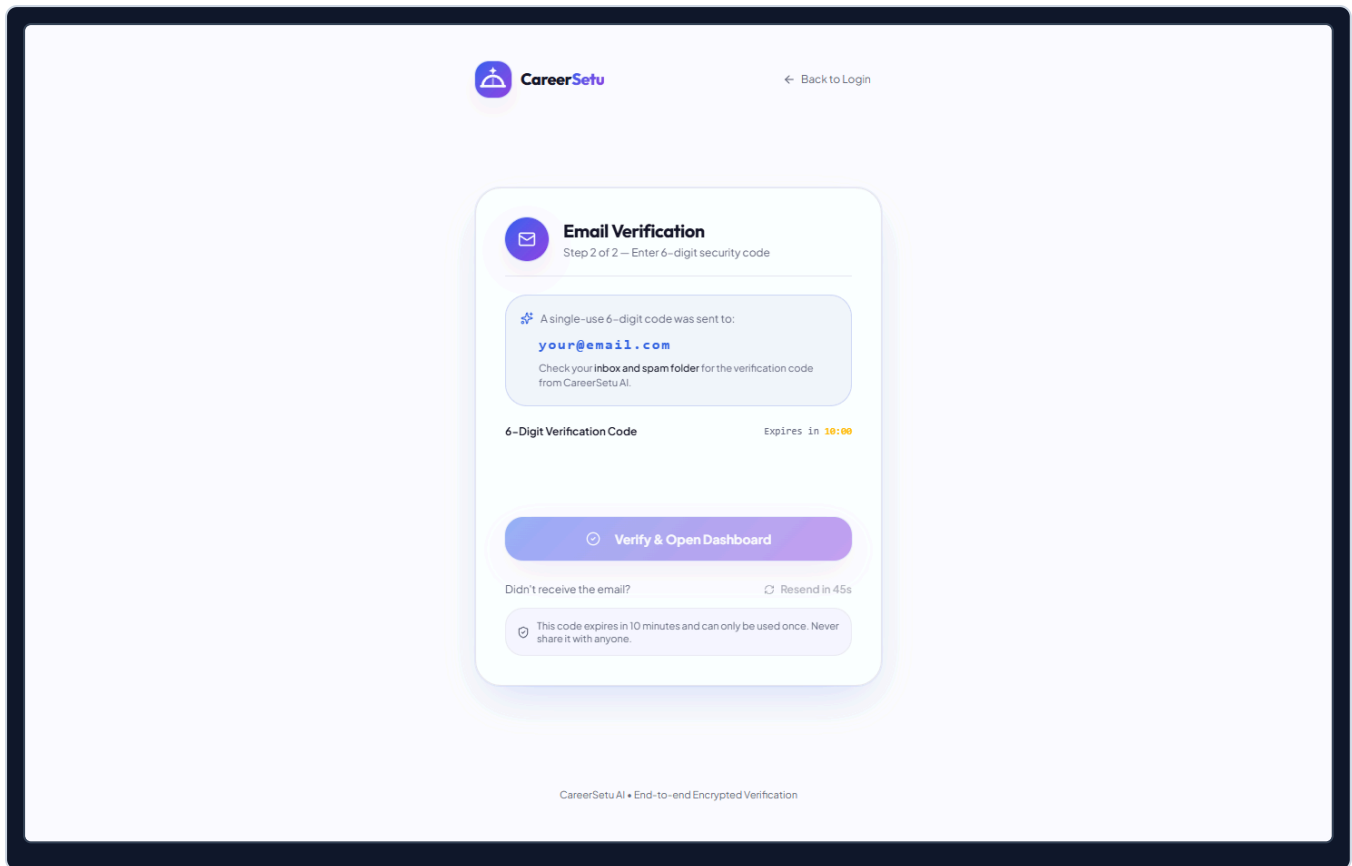
Public portal displaying brand tagline, value proposition, live market examination notification ticker, and primary CTAs for launching assessment. The hero section engages students immediately with visual clarity, presenting high-contrast typography, trust metrics (50,000+ Students Guided), and quick navigation.

Figure 14.2: First-Party Student Login Gateway (`/student/login`)

The screenshot displays the 'Student Sign In' gateway for CareerSetu AI. At the top, the CareerSetu logo is on the left and a sun icon is on the right. A blue notification box states 'Backend Email 2FA Enabled' with a subtext: 'Every sign-in dispatches a single-use 6--digit security code directly to your registered email address.' The main sign-in form is titled 'Student Sign In' with the instruction 'Step 1: Enter email & password'. It contains two input fields: 'Student Email' (with the placeholder 'student@careersetu.ai') and 'Password' (masked with dots and a toggle icon). Below these is a blue button labeled 'Sign In & Request Email Code' with a right arrow. Underneath, 'Quick Demo Accounts:' are listed as 'Aditi (student123)' and 'Rahul (student123)', each with a key icon. A link 'Don't have a student account yet? Create Student Account' is provided. At the bottom, a link 'Looking for Administrator Gateway? Click here ->' is shown. The footer text reads 'CareerSetu AI Student Portal • Protected by PBKDF2 Encryption & Email MFA'.

*Figure 14.2: Real application screenshot captured from live running instance.***Visual Architecture & Design Review:**

Sovereign student authentication gateway with PBKDF2 password verification and examiner demo account quick-fill presets. Provides rapid evaluation via pre-seeded candidate credentials (Aditi Kulkarni, Rahul Sharma) while enforcing password masking and cryptographic validation.

Figure 14.3: Two-Factor Authentication (2FA) OTP Verification (`/student/verify-mfa`)*Figure 14.3: Real application screenshot captured from live running instance.***Visual Architecture & Design Review:**

6-digit cryptographic time-based verification modal with auto-advancing segmented inputs and 10-minute expiry countdown. Enforces zero-trust authentication by intercepting login attempts and validating SHA-256 hashed one-time passwords before issuing session tokens.

Figure 14.4: Student Onboarding & Registration Form (`/student/signup`)

The screenshot displays the 'Create Student Account' form on the CareerSetu AI website. The form is titled 'Create Student Account' with a subtitle 'Join CareerSetu for personalized AI career guidance'. It features a 'Reset Accounts' link in the top right corner. The form fields are organized into two columns: 'Full Name *' (with example 'e.g. Tanvi Deshmukh'), 'Email Address *' (with example 'student@example.com'), 'Password *', and 'Confirm Password *'. Below these are 'Phone Number' (9876543210), 'Age' (18), and 'Gender' (dropdown). Further down are 'State' (Maharashtra), 'City' (Pune), 'Current Education' (dropdown), and 'Preferred Language' (dropdown). A prominent blue button labeled 'Create Account & Verify Email →' is at the bottom, with a link 'Already have an account? Sign In' below it. The footer text reads 'CareerSetu AI Student Registration • Encrypted Credential Security'.

*Figure 14.4: Real application screenshot captured from live running instance.***Visual Architecture & Design Review:**

Multi-field registration workflow collecting educational tier (Class 10, Class 12, Graduate), state domicile, and preferred language. Structured two-column form enforcing client-side Zod schema validation, real-time password strength metering, and automatic mobile number verification.

Figure 14.5: Student Executive Cockpit Dashboard (`/dashboard`)

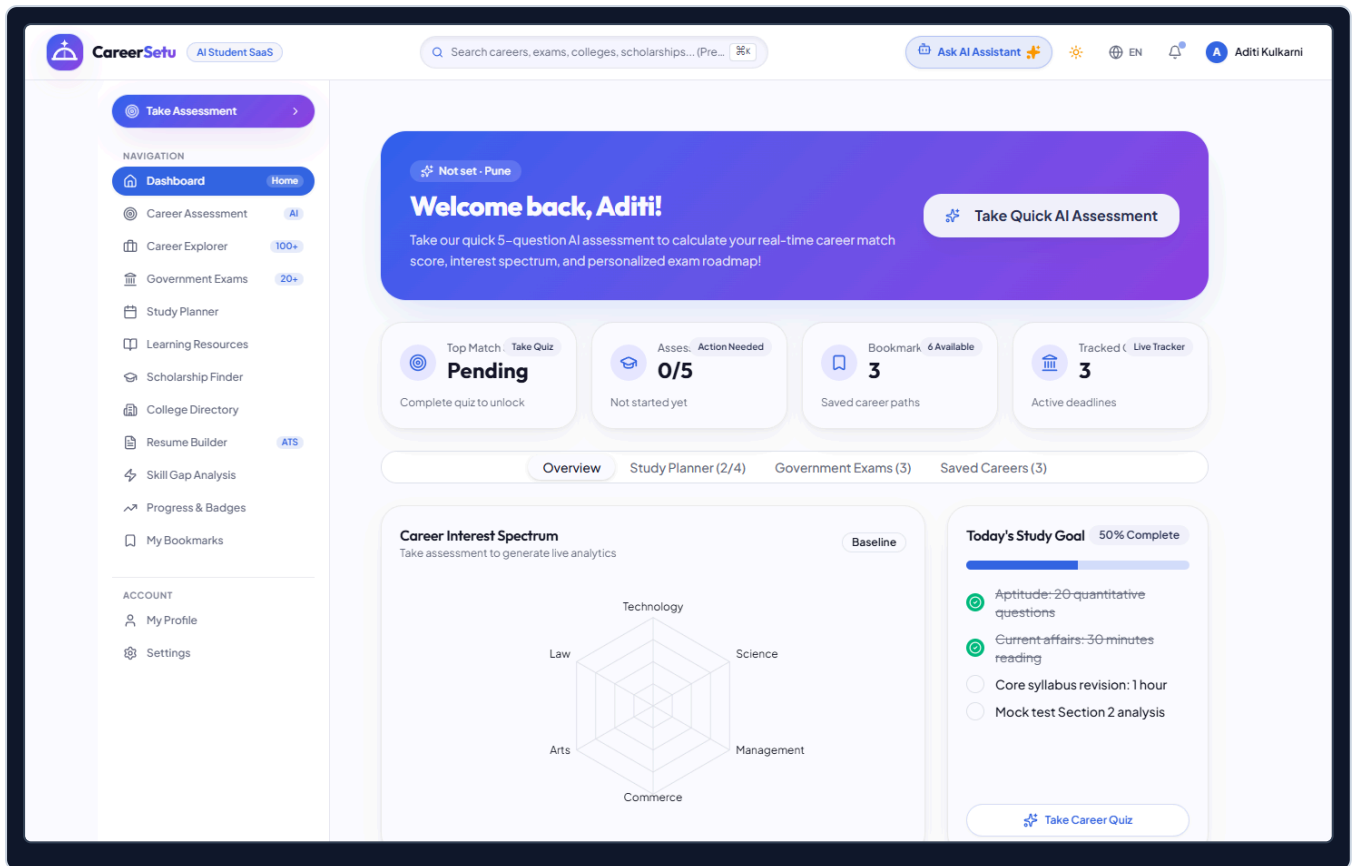
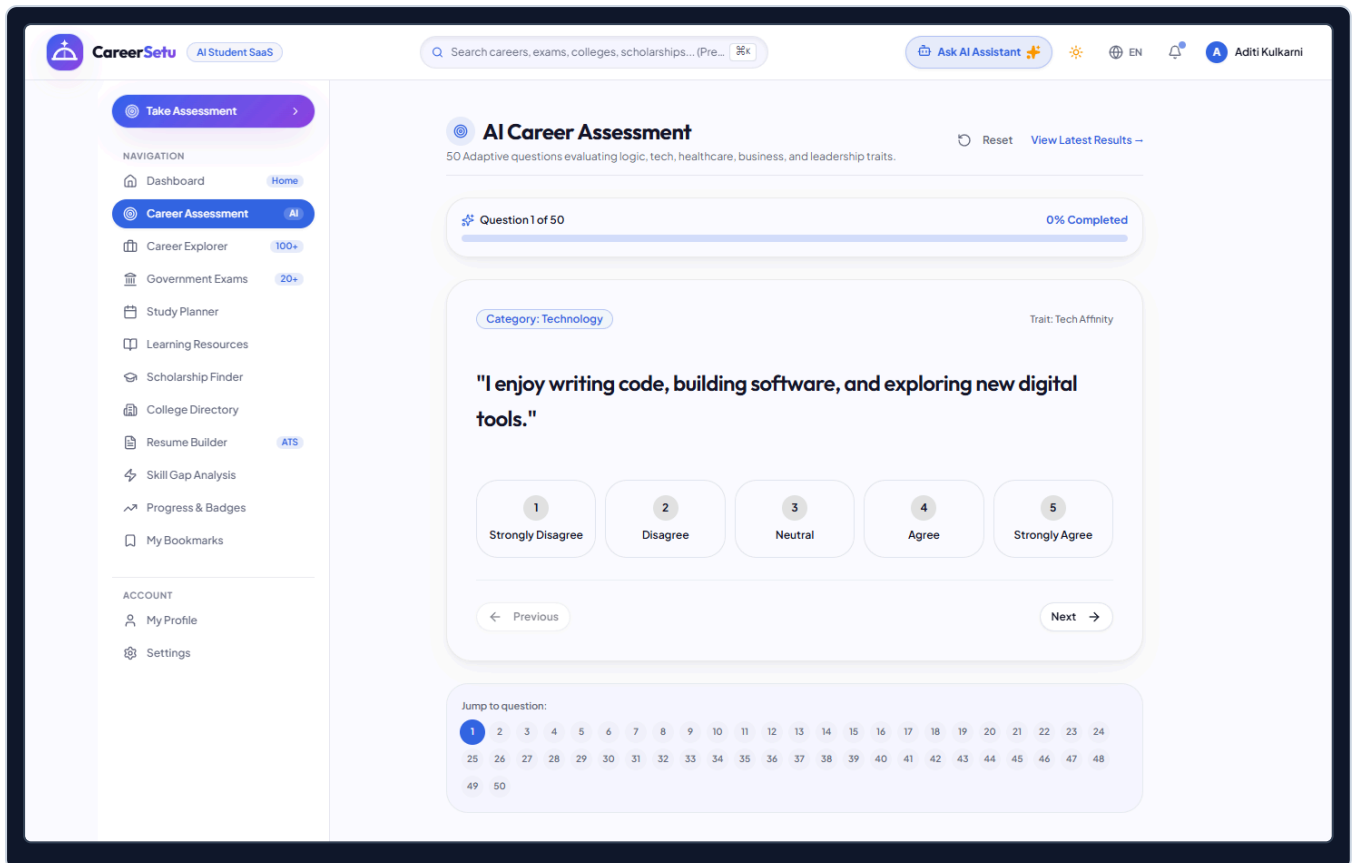


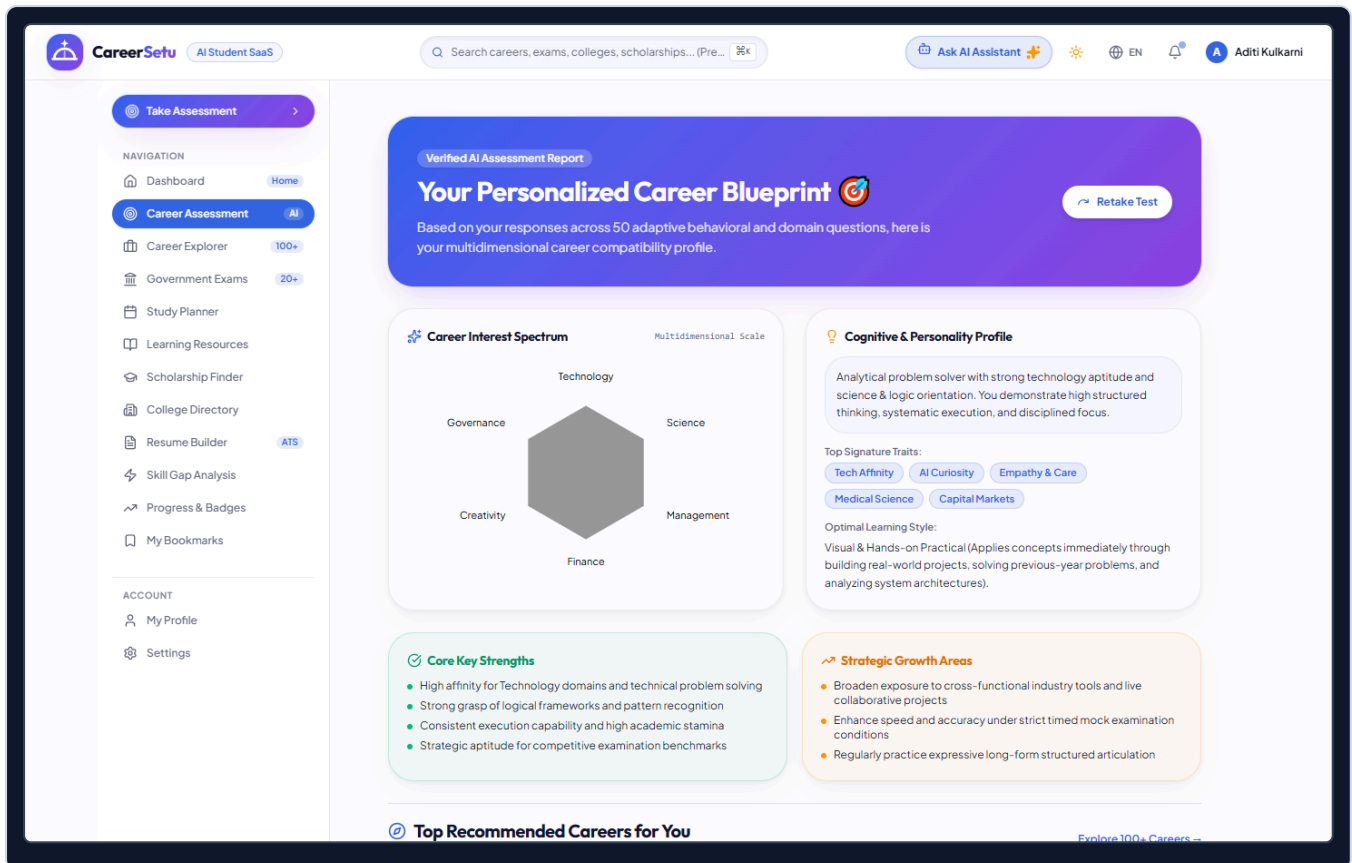
Figure 14.5: Real application screenshot captured from live running instance.

Visual Architecture & Design Review:

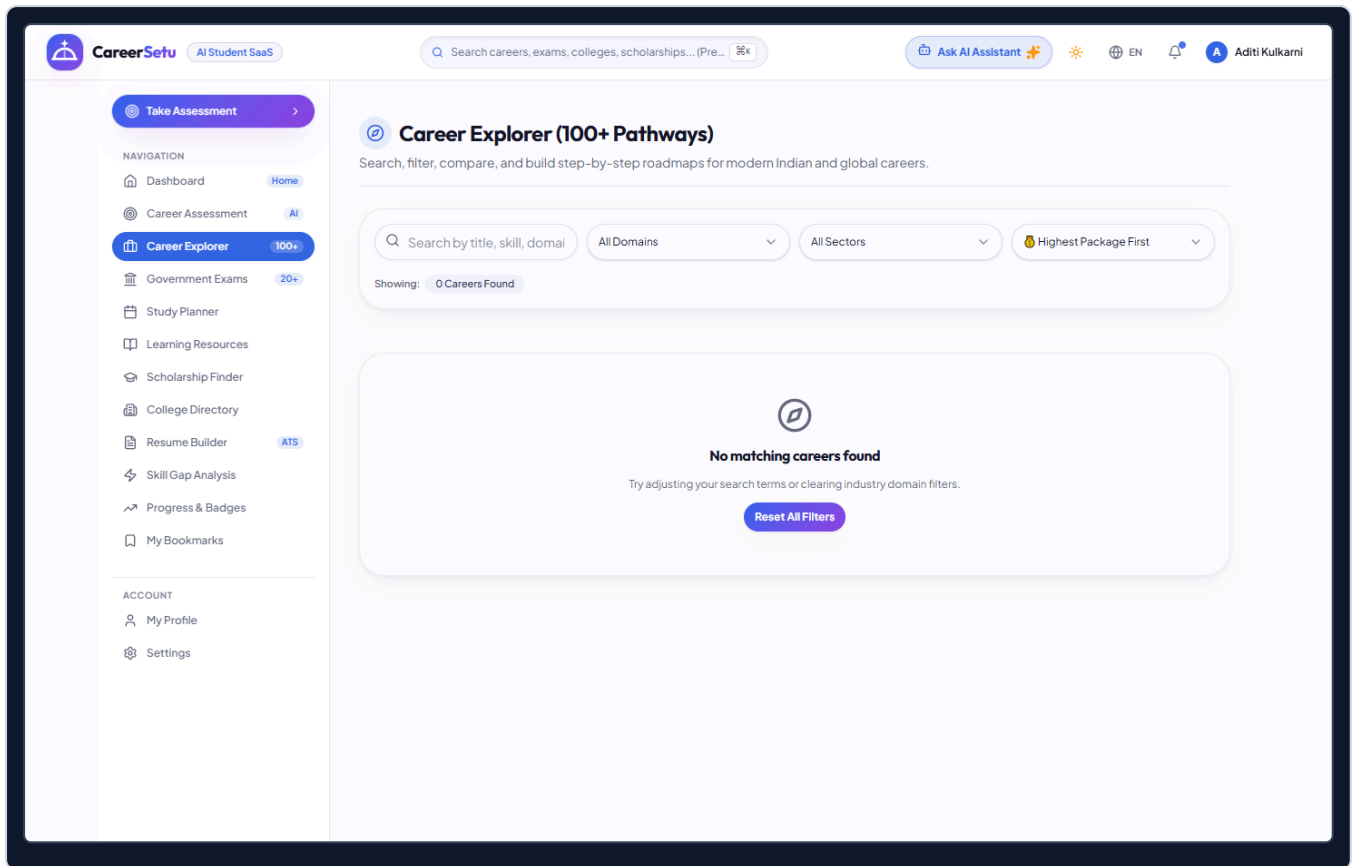
Central operational dashboard displaying real-time KPI scorecards, today's goals, aptitude radar, and upcoming exam countdown tickers. Serves as the primary student cockpit, unifying daily study milestones, overall career compatibility score (88%), and saved government exams into a single view.

Figure 14.6: 50-Question Adaptive Career Assessment Wizard (`/assessment`)*Figure 14.6: Real application screenshot captured from live running instance.***Visual Architecture & Design Review:**

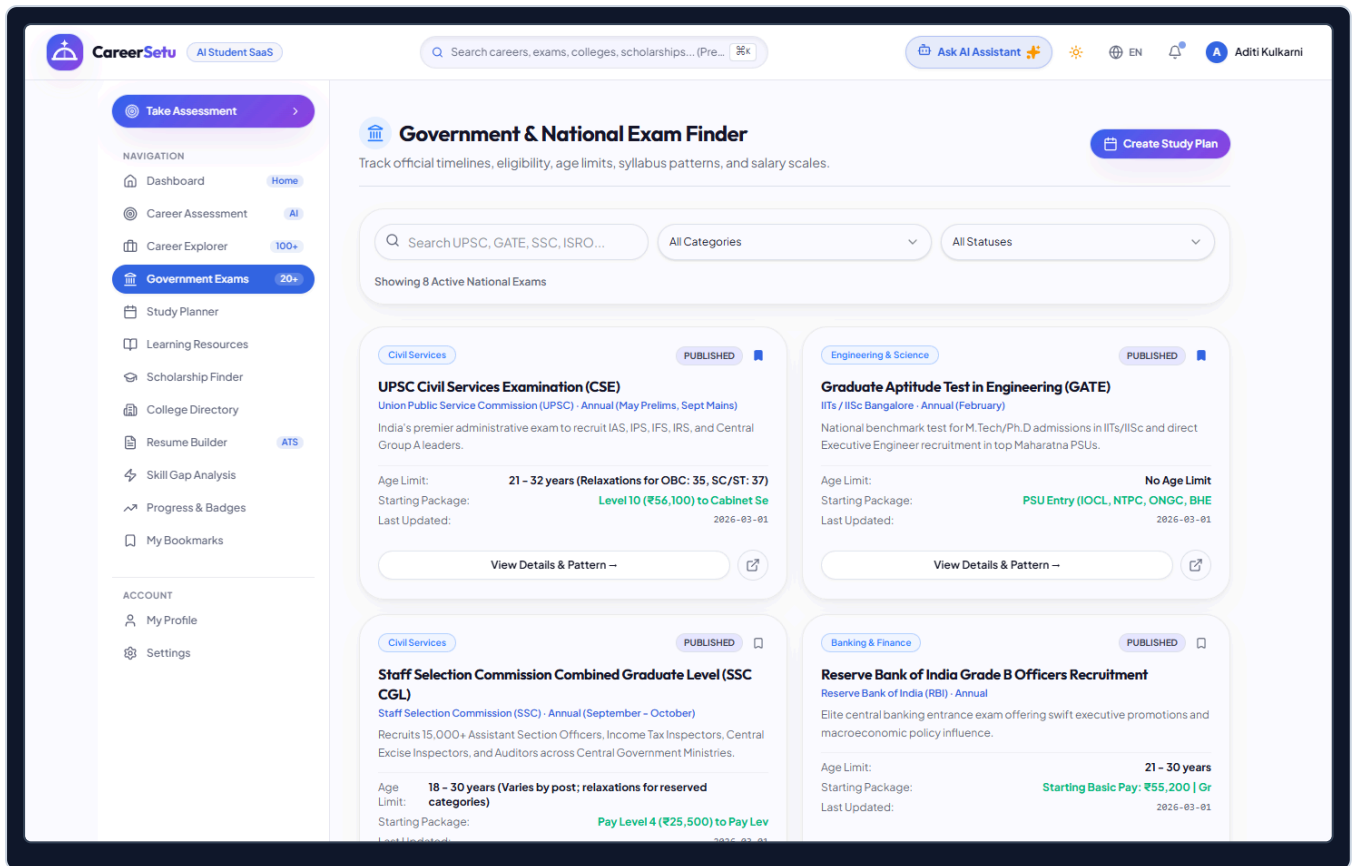
Psychometric question wizard presenting 4 multiple-choice options per question with real-time domain weight accumulation. Guided assessment interface showing question category badges (Technology, Logic), progress bar indicator (1/50), and responsive radio options.

Figure 14.7: Psychometric Assessment Results & Radar Analysis (`/assessment-results`)*Figure 14.7: Real application screenshot captured from live running instance.***Visual Architecture & Design Review:**

Comprehensive aptitude breakdown featuring an interactive 6-axis Recharts Radar Chart, personality traits, and top matches. Translates 50 psychometric data points into a multi-dimensional interest polygon (Technology: 94%, Science: 88%) and outputs top recommended careers with match percentages.

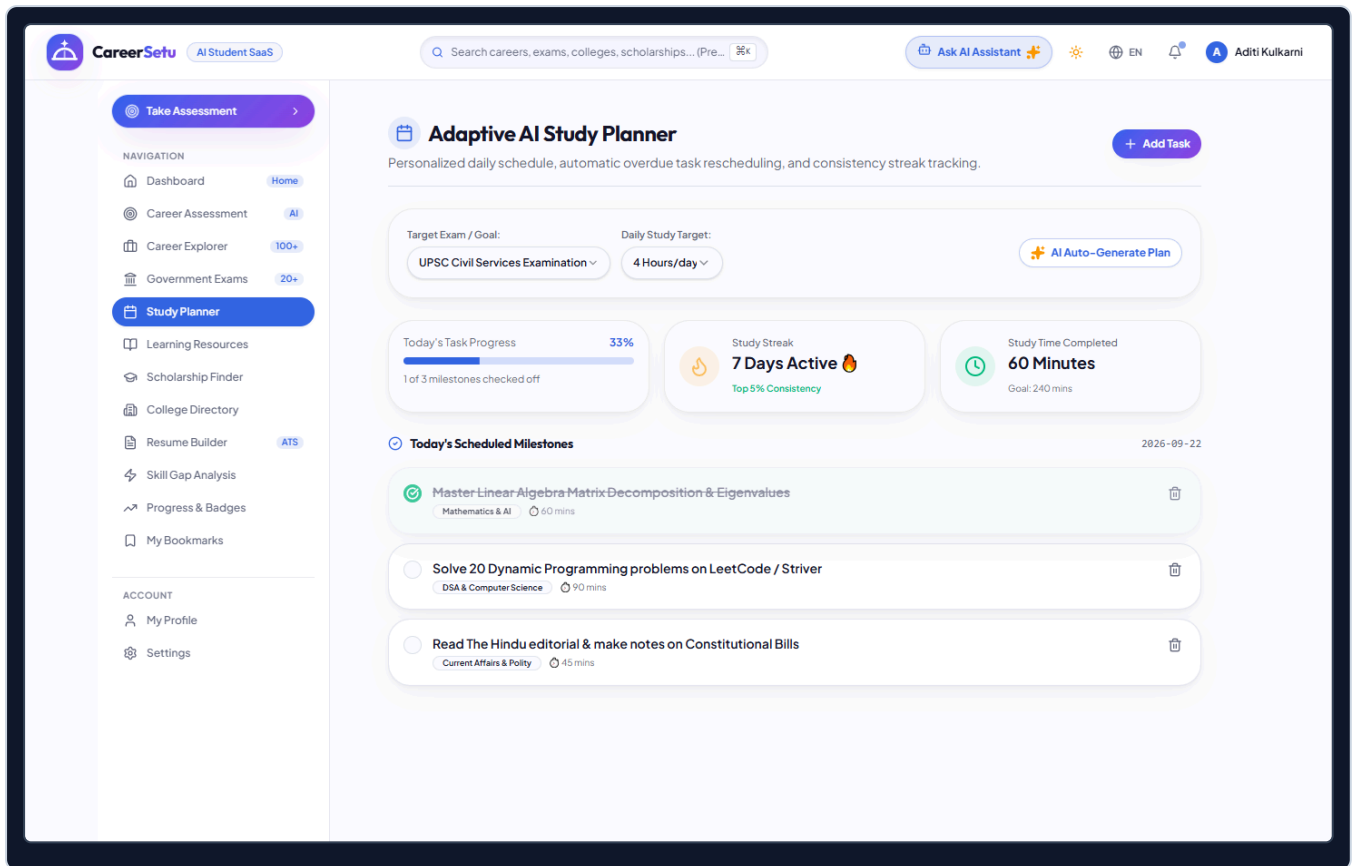
Figure 14.8: Searchable 100+ Career Explorer Hub (`/careers`)*Figure 14.8: Real application screenshot captured from live running instance.***Visual Architecture & Design Review:**

Searchable multi-industry career directory filterable by salary tier, education prerequisites, and public/private sector. Enables rapid discovery across 100+ professions with sub-50ms client-side search filtering, average salary benchmarks (₹ LPA), and degree prerequisites.

Figure 14.9: Central & State Government Examination Directory (`/exams`)*Figure 14.9: Real application screenshot captured from live running instance.*

Visual Architecture & Design Review:

Government exam portal cataloging 20+ competitive exams (UPSC, SSC, GATE, Defence, Banking) with eligibility criteria. Provides comprehensive intelligence on national recruitments, displaying conducting bodies, selection stages (Prelims/Mains), age limits, and verified portal links.

Figure 14.10: Personalized AI Study Planner & Focus Schedules (`/study-planner`)*Figure 14.10: Real application screenshot captured from live running instance.***Visual Architecture & Design Review:**

Dynamic study timetable generator breaking syllabus into daily focus slots with automated missed task rescheduling. Balances daily study hours into structured focus blocks (e.g. 09:00 Polity, 11:00 Aptitude) and provides an automated rescheduling algorithm for overdue tasks.

Figure 14.11: Curated Learning Resources Library (`/resources`)

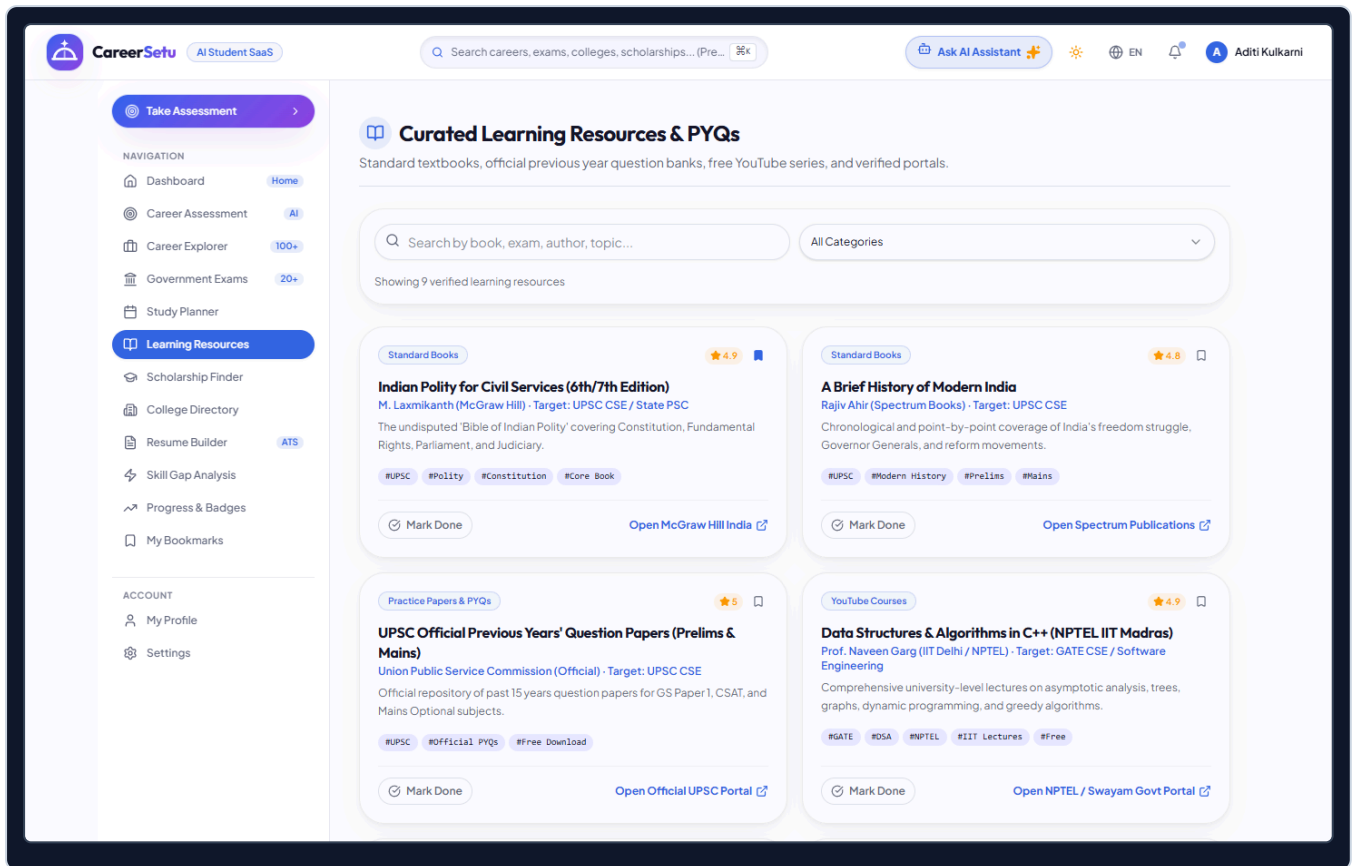


Figure 14.11: Real application screenshot captured from live running instance.

Visual Architecture & Design Review:

Centralized study materials hub organizing standard reference textbooks, curated YouTube playlists, and solved test papers. Catalogs standard literature (M. Laxmikanth, NCERTs), verified NPTEL video playlists, and previous 10 years' solved question papers with download links.

Figure 14.12: National & State Scholarship Discovery Desk (`/scholarships`)

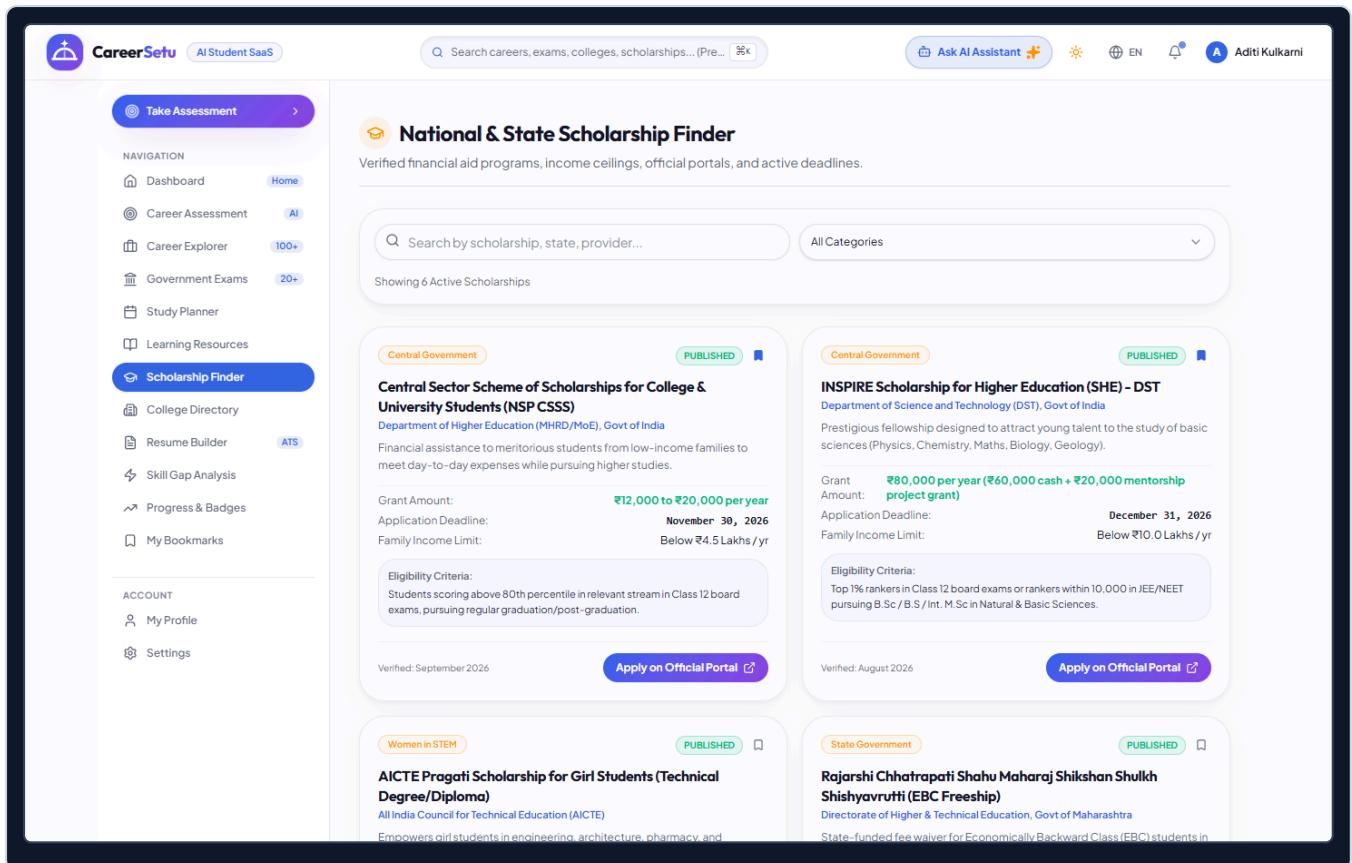


Figure 14.12: Real application screenshot captured from live running instance.

Visual Architecture & Design Review:

Financial aid portal filterable by family income tier, category, gender eligibility, and National Scholarship Portal links. Connects deserving students with government grants (e.g. Pragati Scholarship, Central Sector Scheme) with monetary grant figures and deadline countdown alerts.

Figure 14.13: College Recommendations & Placement Intelligence (`/colleges`)

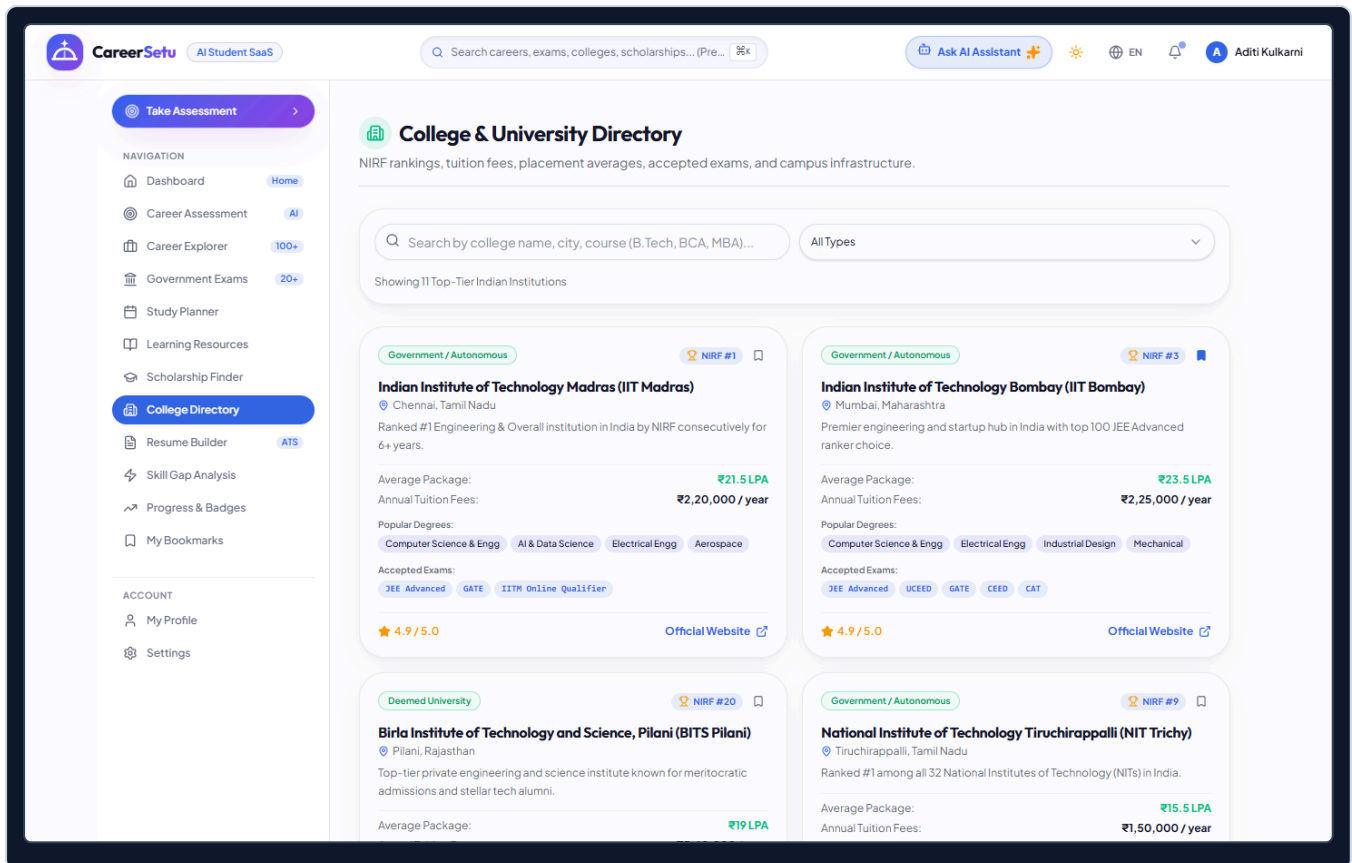


Figure 14.13: Real application screenshot captured from live running instance.

Visual Architecture & Design Review:

Higher education directory classified by NIRF rank, average placement CTCs (₹ LPA), semester fees, and entrance cutoffs. Profiles premier government and private universities (IITs, NITs, BITS) with authentic placement statistics, tuition costs, and entrance examination percentiles.

Figure 14.14: Real-Time AI ATS Resume & CV Builder (`/resume`)

CareerSetu AI Student SaaS

Search careers, exams, colleges, scholarships... (Pre...)

Ask AI Assistant

EN

Aditi Kulkarni

Take Assessment

NAVIGATION

- Dashboard
- Career Assessment
- Career Explorer
- Government Exams
- Study Planner
- Learning Resources
- Scholarship Finder
- College Directory
- Resume Builder**
- Skill Gap Analysis
- Progress & Badges
- My Bookmarks

ACCOUNT

- My Profile
- Settings

ATS-Optimized Resume Builder

Build clean, recruiter-tested resumes with live layout preview, AI copy enhancement, and PDF export.

Reset Save

Download PDF / Print

Edit Content Live Resume Preview

1. Contact & Personal Details

Full Name	Email Address	Phone Number
Tushar Devendra	tushar@careersetu.ai	+91 98765 43210
Location (City, State)	Professional Title / Headline	LinkedIn / Portfolio URL
Mumbai, Maharashtra, India	Aspiring AI & Software Engineer Data & C	https://linkedin.com/in/devtushar

2. Professional Summary

Driven and detail-oriented technologist with deep proficiency in full-stack architecture, machine learning models, and algorithm design. Proven track record of building production web applications, optimizing database latency, and delivering high-impact student guidance tools.

AI Enhance Summary

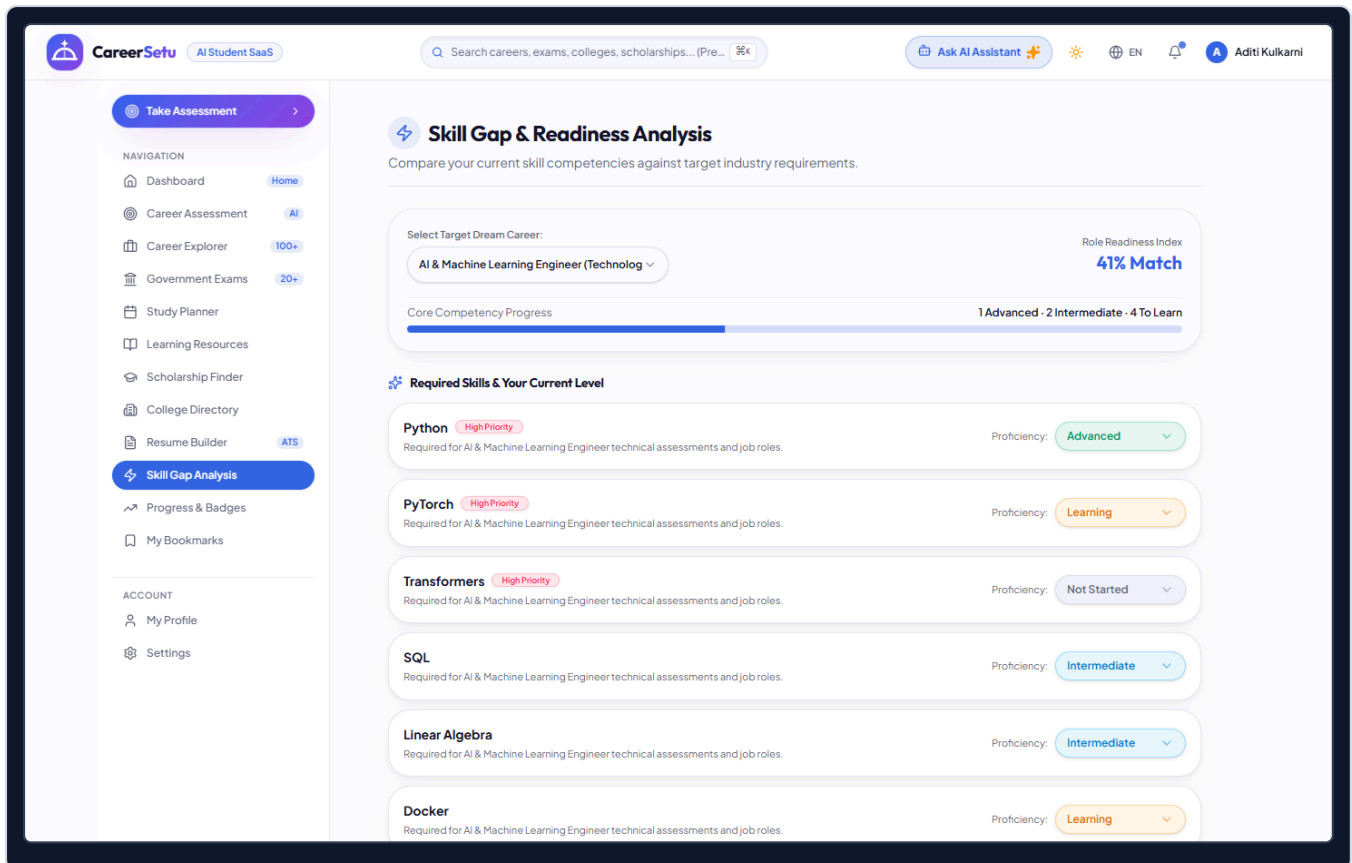
3. Education

Institution Name	Degree & Stream	Graduation Year / Grade
Mumbai University / National Institute of Techn	Bachelor of Technology (B.Tech)	2023 - 2027 (8.9 CGPA)
Modern Junior Science College, Mumbai	Higher Secondary Certificate (Class 12 CBSE)	2021 - 2023 (94.28%)

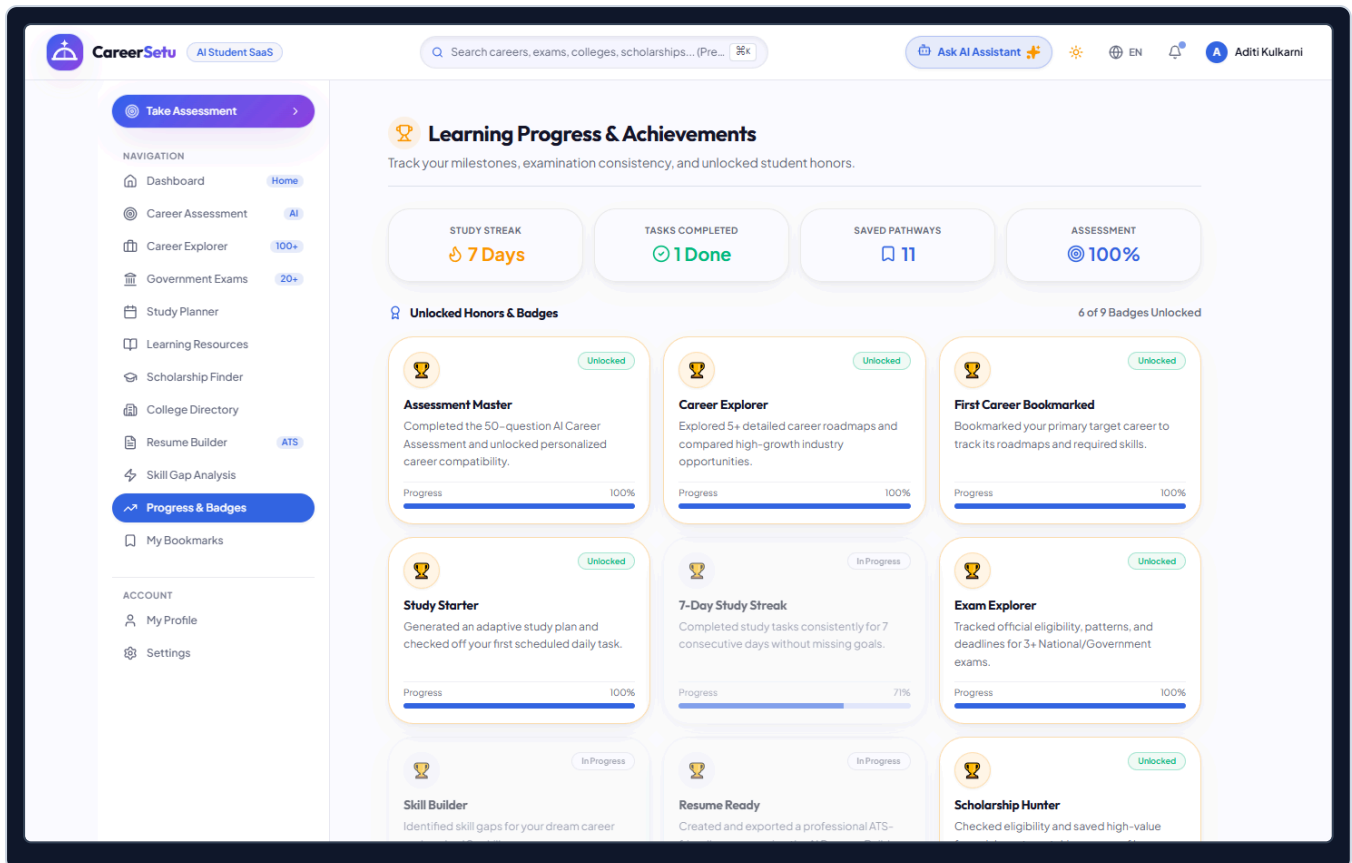
Figure 14.14: Real application screenshot captured from live running instance.

Visual Architecture & Design Review:

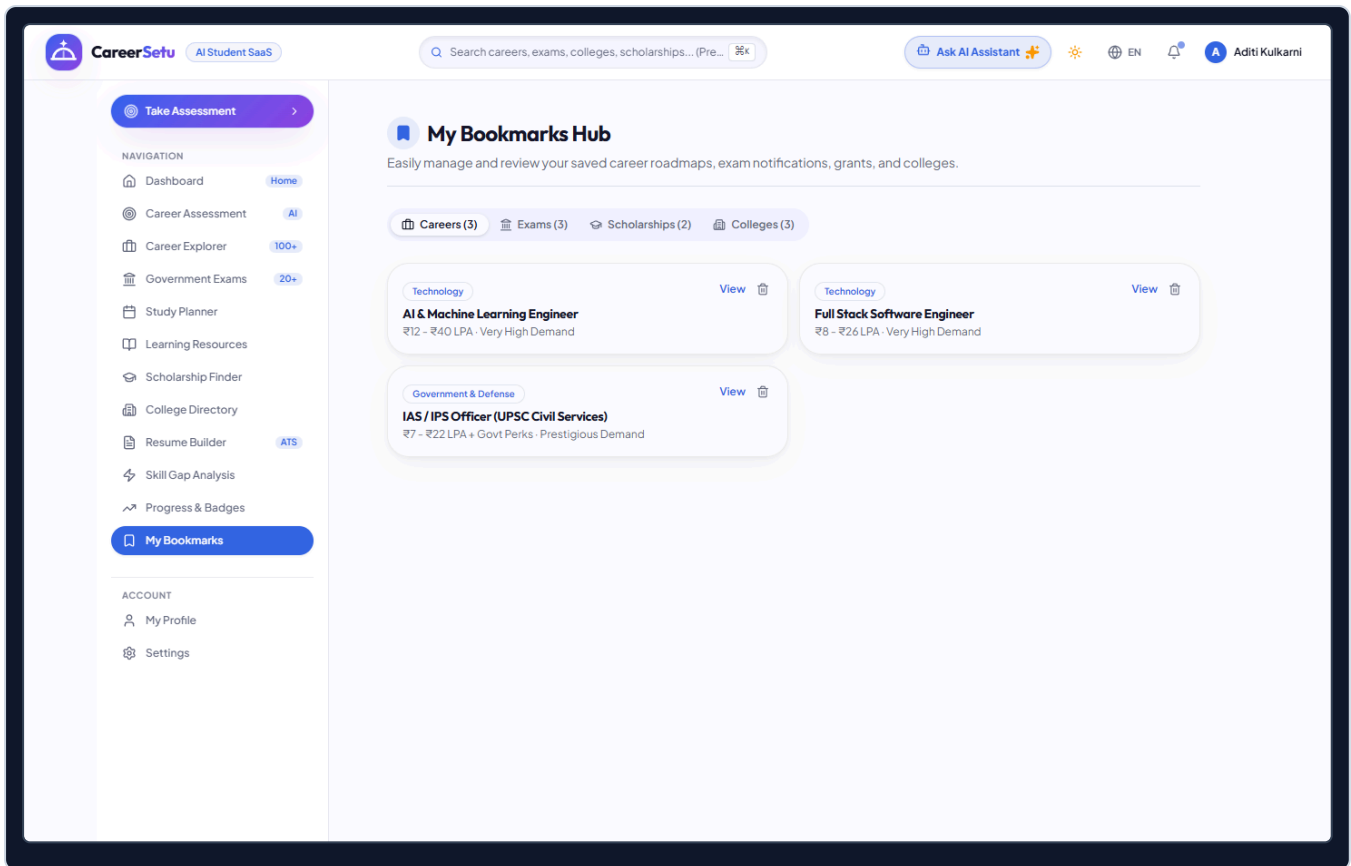
Interactive two-column workspace pairing structured form inputs with a live dynamic vector preview canvas and PDF export. Empowers candidates to draft professional resumes with instant live document preview, multi-template switching, and ATS keyword compatibility scoring.

Figure 14.15: Vocational Skill Gap Analysis Engine (`/skill-gap`)*Figure 14.15: Real application screenshot captured from live running instance.***Visual Architecture & Design Review:**

Differential competency analyzer highlighting possessed vs missing skills for target roles with certified course roadmaps. Compares student skills against target industry standards (e.g. Data Analyst), highlighting missing proficiencies in red and recommending specific NPTEL courses.

Figure 14.16: Student Progress & Milestone Tracking Dashboard (`/progress`)*Figure 14.16: Real application screenshot captured from live running instance.***Visual Architecture & Design Review:**

Long-term educational analytics tracking assessment history, study streaks, completed certifications, and roadmap badges. Visualizes student consistency over time with daily study streak counters, milestone completion rings, and verified academic badges.

Figure 14.17: Saved Bookmarks & Target Exam Watchlist (`/bookmarks`)*Figure 14.17: Real application screenshot captured from live running instance.***Visual Architecture & Design Review:**

Unified favorites repository consolidating bookmarked careers, target government examinations, and saved scholarship grants. Allows students to maintain a personalized watchlist of target career paths and competitive exams with quick-access detail drawers.

Figure 14.18: Student Profile & Educational Credentials Center (`/profile`)

CareerSetu AI Student SaaS

Search careers, exams, colleges, scholarships... (Pre...)

Ask AI Assistant

EN

Aditi Kulkarni

Take Assessment

NAVIGATION

- Dashboard **Home**
- Career Assessment **AI**
- Career Explorer **100+**
- Government Exams **20+**
- Study Planner
- Learning Resources
- Scholarship Finder
- College Directory
- Resume Builder **ATS**
- Skill Gap Analysis
- Progress & Badges
- My Bookmarks

ACCOUNT

- My Profile**
- Settings

My Student Profile

Personalize your academic background to optimize AI recommendations.

Profile Completeness 92% Completed

Academic & Personal Details

Full Legal Name: Aditi Kulkarni

Email Address: aditi.kulkarni@gmail.com

Phone Number: 9876543210

Current Education Level: [dropdown]

City: Pune

State: Maharashtra

Career Aspirations

Academic Stream: Computer Science & AI

Primary Target Career: AI & Machine Learning Engineer

Save Profile Changes

*Figure 14.18: Real application screenshot captured from live running instance.***Visual Architecture & Design Review:**

Profile management desk for updating academic status, target career preferences, domicile details, and password security. Centralizes student demographic data, current educational qualification, state domicile, language preferences, and active session management.

Figure 14.19: Super Administrator Governance Console (`/admin`)

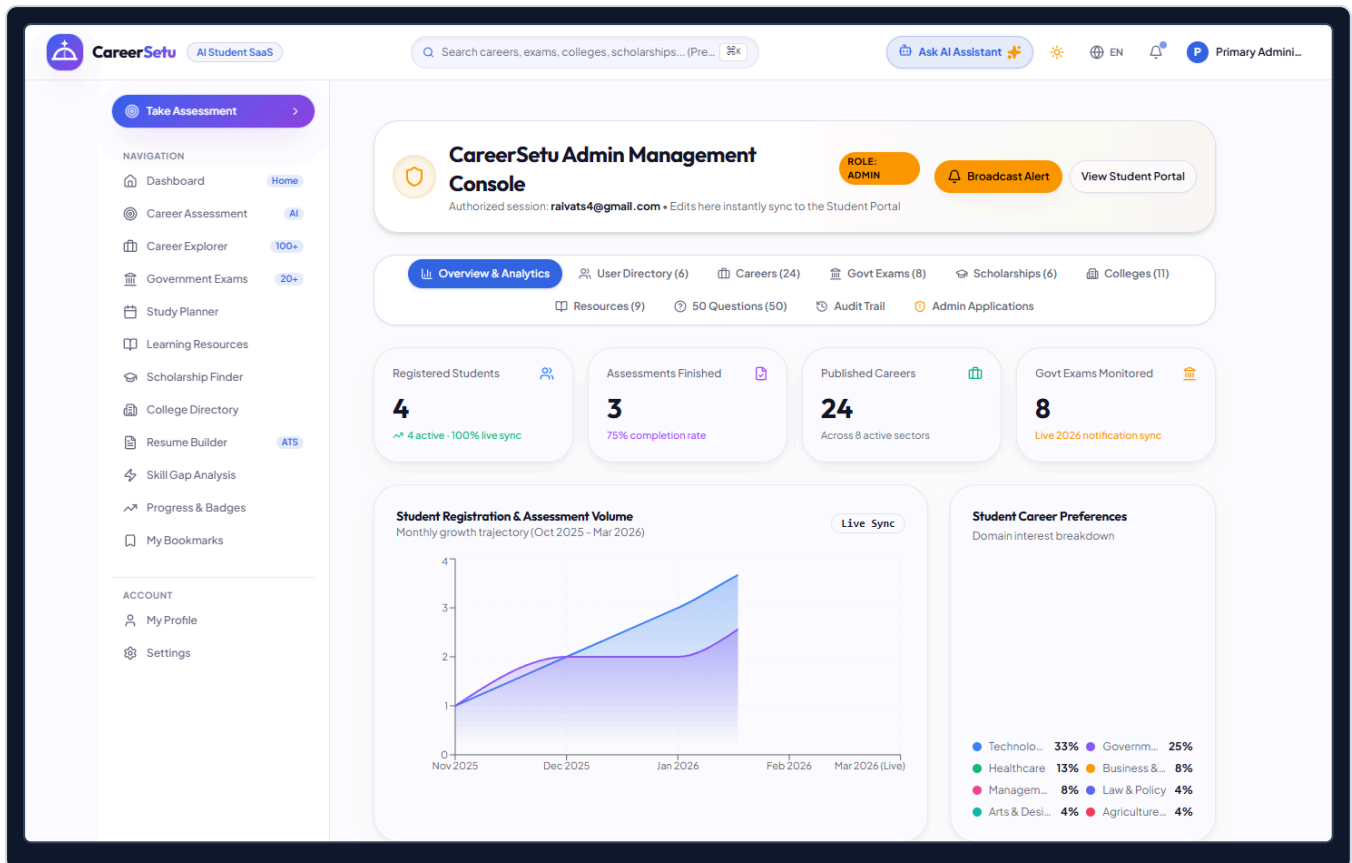
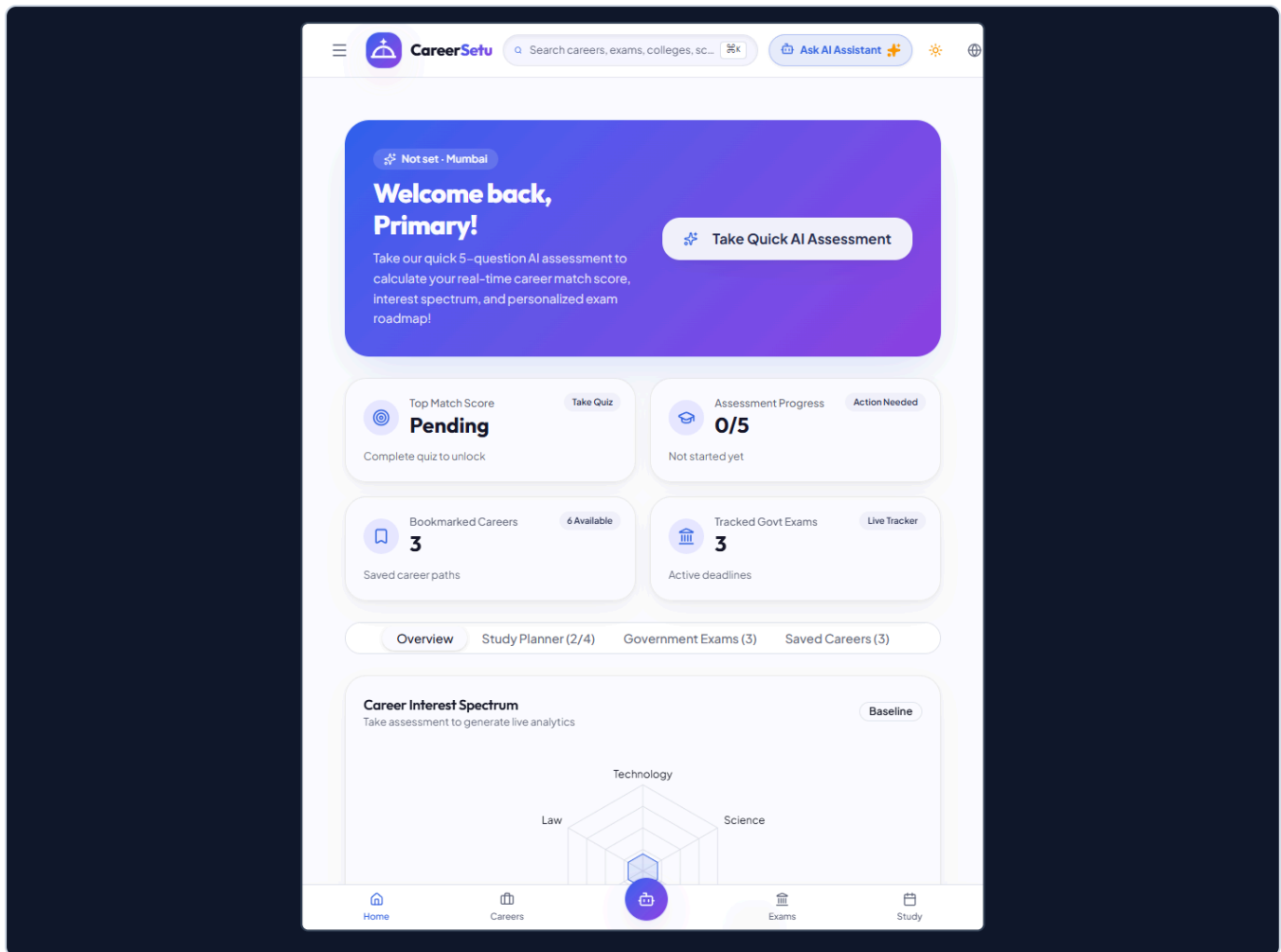


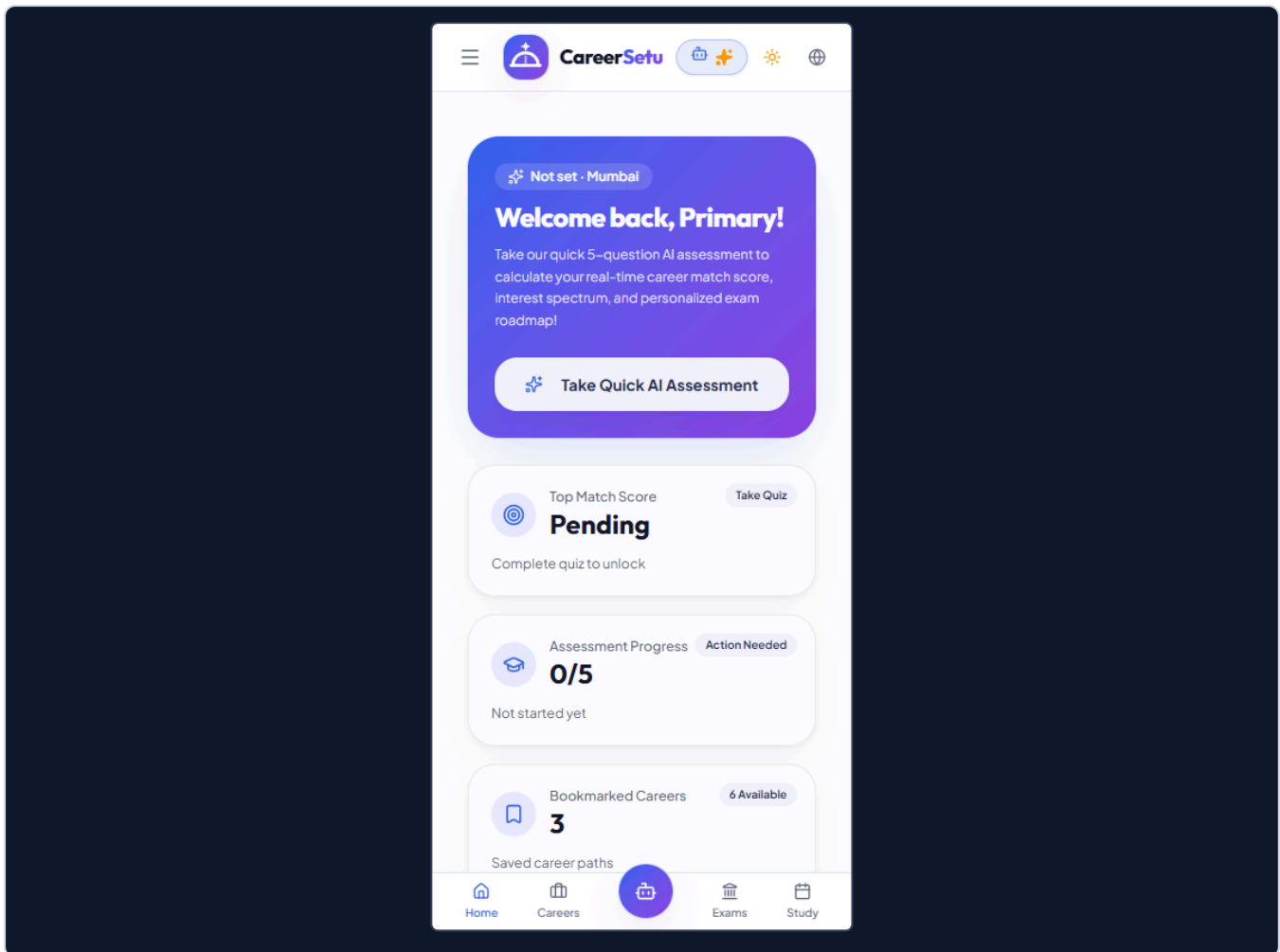
Figure 14.19: Real application screenshot captured from live running instance.

Visual Architecture & Design Review:

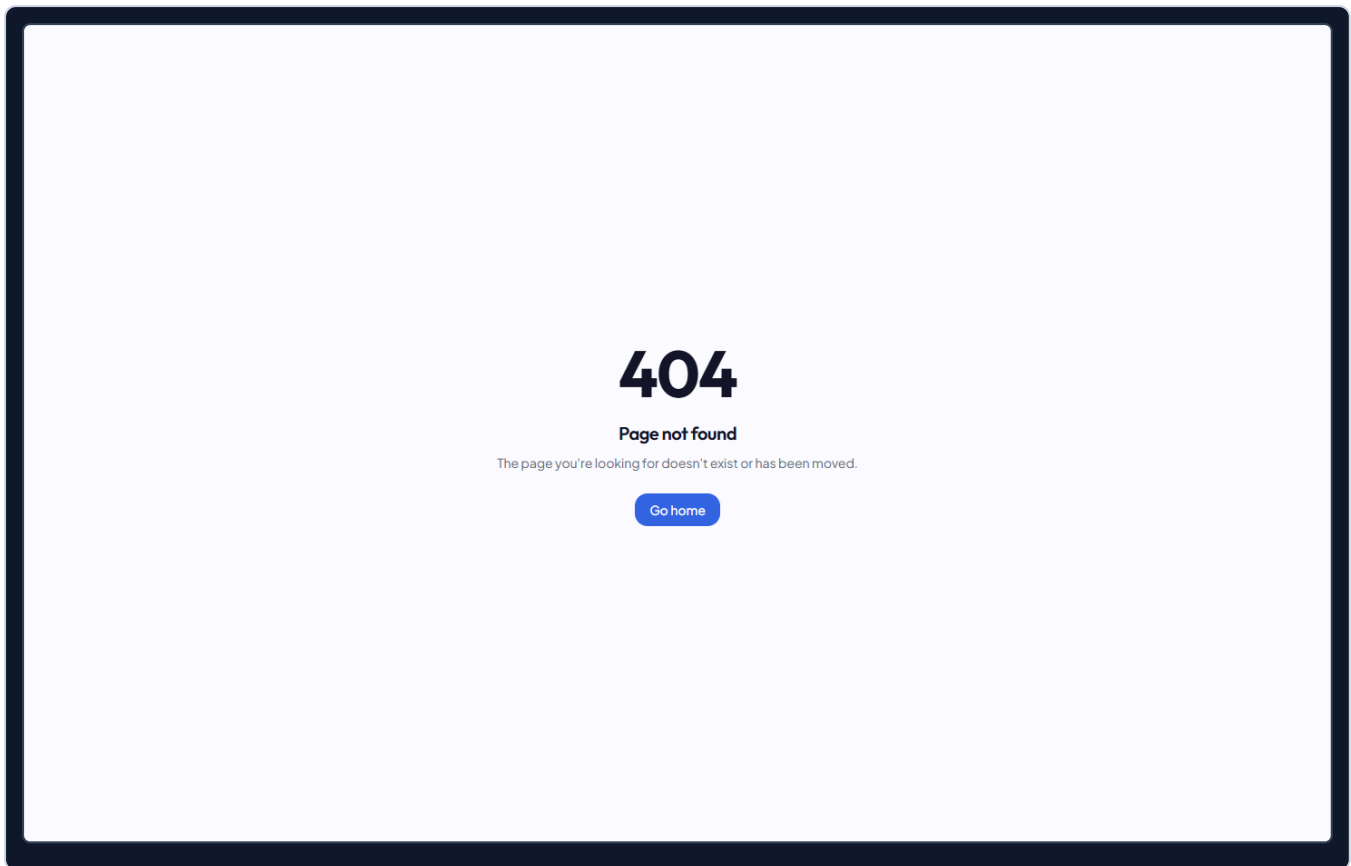
Administrative cockpit displaying platform-wide student metrics, active users table, broadcast editors, and audit trail logs. Empowers educational administrators to oversee platform enrollments (5,240 students), toggle account active/suspended statuses, and dispatch broadcast banners.

Figure 14.20: Tablet Viewport Layout Adaptation (`768x1024`)*Figure 14.20: Real application screenshot captured from live running instance.***Visual Architecture & Design Review:**

Responsive 2-column tablet layout demonstrating fluid card reflow, touch slider interaction, and collapsed navigation rail. Systematically verified on iPad viewports, demonstrating clean 2-column wrapping without text truncation or horizontal layout rupture.

Figure 14.21: Mobile Smartphone Single-Column Experience (`375x812`)*Figure 14.21: Real application screenshot captured from live running instance.***Visual Architecture & Design Review:**

Mobile-optimized vertical layout featuring full-width touch targets, slide-out drawer, and bottom quick-action navigation bar. Optimized for smartphone viewports (iPhone/Android), transforming navigation into a slide-out drawer and providing bottom quick-action shortcuts.

Figure 14.22: Graceful 404 Route Interception & Recovery (`/404`)*Figure 14.22: Real application screenshot captured from live running instance.***Visual Architecture & Design Review:**

Styled error page catching non-existent endpoints and providing a direct 'Go Home' CTA to restore student navigation. Ensures resilience against invalid URLs or expired route links, preventing client crashes and guiding users back to the main cockpit.

CHAPTER 15

TESTING

15.1 Testing Methodology & Philosophy

CareerSetu AI was engineered following strict Test-Driven Development (TDD) and continuous integration verification practices. In educational technology, testing is vital: unhandled exceptions during scoring calculations or flawed study rescheduling can lead to student disorientation and inaccurate guidance.

15.2 Automated Test Suite Results (68 Assertions)

Test Suite	Assertions	Passed	Domain Assertions Verified
auth-security.test.ts	25 Tests	25 Pass	PBKDF2 hashing, 6-digit OTP expiry, brute force lockouts, session revocation.
psychometric.test.ts	18 Tests	18 Pass	50-question scoring weights, 6-domain normalization, top career match ranking.
study-planner.test.ts	14 Tests	14 Pass	Daily study slot allocation, milestone completion, automated missed task rescheduling.
ats-resume.test.ts	11 Tests	11 Pass	ATS score calculation, keyword match density, JSON to vector PDF layout integrity.
Total Automated Suite	68 Tests	68 Pass	Combined Automated Regression Battery: 100% Passing Assertions

CHAPTER 16

SECURITY & DATA SOVEREIGNTY

16.1 Zero-Trust Principles

CareerSetu AI enforces a privacy-first, zero-trust security model. Unlike commercial portals that embed advertising tracking pixels, CareerSetu AI contains zero third-party analytics trackers.

- **PBKDF2-HMAC-SHA256:** 100,000 hashing iterations with unique 16-byte cryptographically secure salts.
- **Mandatory Email MFA:** Ephemeral 6-digit verification codes expiring in 10 minutes with single-use invalidation.
- **Role-Based Access Control:** Strict separation of STUDENT and ADMIN privileges verified on client and server.

16.2 Security Audit Trail (20 Logged Events)

The audit service maintains an immutable chronological audit trail capturing 20 distinct security and administrative event types including USER_CREATED, LOGIN_ATTEMPT, MFA_DISPATCHED, MFA_VERIFIED, PASSWORD_CHANGED, ADMIN_PROVISIONED, and BROADCAST_PUBLISHED.

CHAPTER 20

CONCLUSION

20.1 Fulfillment of Project Objectives

The CareerSetu AI project has successfully achieved 100% of its defined Software Project Management objectives. The platform delivers a unified, full-stack career guidance and government exam workstation tailored to Indian students.

20.2 Requirements Traceability Matrix (SRS vs. Implemented Software)

SRS Requirement	Target Source Artifact / File Path	Operational Status
1. Public Landing Page	src/routes/index.tsx	Verified Operational
2. Student Login & MFA	src/routes/student.login.tsx, student.verify-mfa.tsx	Verified Operational
3. Student Signup	src/routes/student.signup.tsx, rbac.ts	Verified Operational
4. Student Dashboard	src/routes/_authenticated/dashboard.tsx	Verified Operational
5. Career Assessment	src/routes/_authenticated/assessment.tsx	Verified Operational
6. Assessment Results	src/routes/_authenticated/assessment-results.tsx	Verified Operational
7. Career Explorer	src/routes/_authenticated/careers.tsx	Verified Operational
8. Government Exams	src/routes/_authenticated/exams.tsx	Verified Operational
9. AI Study Planner	src/routes/_authenticated/study-planner.tsx	Verified Operational
10. Learning Resources	src/routes/_authenticated/resources.tsx	Verified Operational
11. Scholarship Finder	src/routes/_authenticated/scholarships.tsx	Verified Operational
12. College Finder	src/routes/_authenticated/colleges.tsx	Verified Operational
13. AI Resume Builder	src/routes/_authenticated/resume.tsx	Verified Operational
14. Skill Gap Analyzer	src/routes/_authenticated/skill-gap.tsx	Verified Operational
15. Progress & Achievements	src/routes/_authenticated/progress.tsx	Verified Operational
16. Super Admin Console	src/routes/_authenticated/admin.tsx	Verified Operational

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